MATH 2411: Applied Statistics

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1 Descriptive Statistics

1.1 Data and Variable Types

1.1.1 What are Data?

Data (singular form: datum) in science are a collection of measurements and identifiers. They can take various forms including numerical, character, or other categories. Without summarizing, data are non-informative (e.g., a raw list of 25 exam scores tells us little until summarized).

- **Case / Subject:** A single unit/patient/individual in the sample (each *row* of a data table collects all measurements of one case).
- **Variable:** A specific characteristic of interest corresponding to a *column* in a dataset.
- **Observation:** The value (datum) of a single variable for a single case/individual.

1.1.2 Classification of Variables

Variables are broadly classified into two main categories and four sub-types:

1. **Categorical / Qualitative Variables:** Questions that yield distinct categorical or label responses.
   - **Nominal:** Categories cannot be naturally ranked or ordered.
     Examples: Computer ownership (Yes/No), Birth location, Favorite ice cream flavor.
   - **Ordinal:** Categories possess a natural order or ranking.
     Examples: Instructor evaluation ratings (Very Bad, Bad, Satisfactory, Good, Very Good), Highest level of education, Severity of pain.
2. **Quantitative Variables:** Questions that produce numerical responses.
   - **Discrete:** Arise from a counting process (take integer values, e.g., $0, 1, 2, 3, \dots$).
     Examples: Number of courses enrolled, Children per household, Online courses taught.
   - **Continuous:** Arise from a measuring process (can take any real value in an interval).
     Examples: Height (168.5 cm), Weight, Battery life span.

Why should we care? Knowing the type helps (i) decide how to interpret the data (nominal values are just codes), and (ii) decide what analysis is appropriate. **Important Remark:** Categorical variables are often numerically coded in practice (e.g., yes $= 1$, no $= 0$). Do not mistake these for quantitative variables! Averaging nominal codes is invalid and statistically meaningless.

1.2 Numerical Summaries

Given n observations in a sample labeled by $x_1, x_2, \dots, x_n$:

1.2.1 Measures of Location / Center

1. **Sample Mean:** average of the data, $\boxed{\overline{x} = \frac{1}{n}\sum_{i=1}^{n} x_i = \frac{x_1 + x_2 + \cdots + x_n}{n}}$. *Property:* Sensitive to outliers.
2. **Sample Median ($\tilde{x}$):** The middle point of the ordered data. With data sorted $x_{(1)} \leq x_{(2)} \leq \cdots \leq x_{(n)}$: $\boxed{\tilde{x} = \begin{cases} x_{\left(\frac{n+1}{2}\right)}, & \text{if } n \text{ is odd} \\ \frac{1}{2}\left(x_{\left(\frac{n}{2}\right)} + x_{\left(\frac{n}{2}+1\right)}\right), & \text{if } n \text{ is even} \end{cases}}$ *Property:* Robust and insensitive to outliers.
3. **Trimmed Sample Mean:** Computed by trimming away a certain percent of both the largest and smallest values. E.g., the 10% trimmed mean $\overline{x}_{tr(10)}$ eliminates the largest 10% and smallest 10% and averages the rest. *Properties:* More insensitive to outliers than the sample mean (but not as insensitive as the median), and it uses more information than the sample median. *Remark:* Sample mean, median and trimmed mean can be quite different. *Example (stem weights):* no nitrogen: $\overline{x} = 0.399$, $\tilde{x} = 0.400$, $\overline{x}_{tr(10)} = 0.39750$; nitrogen: $\overline{x} = 0.565$, $\tilde{x} = 0.505$, $\overline{x}_{tr(10)} = 0.56625$ grams.

1.2.2 Measures of Variability / Spread

1. **Sample Variance:** the "average" of the squared deviations from the sample mean. Two common versions: $s_n^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \overline{x})^2$ and $s_{n-1}^2 = \boxed{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \overline{x})^2}$. Unless otherwise specified, this course uses s_{n-1}^2. The quantity $n-1$ is the *degrees of freedom* of the variance estimate (the number of independent pieces of information for computing variability), since $\sum_{i=1}^{n}(x_i - \overline{x}) = 0$ (e.g., data $5, 17, 6, 4$ give $\overline{x} = 8$, and $(-3)^2 + 9^2 + (-2)^2 + (-4)^2$ uses only 3 free deviations).
2. **Sample Standard Deviation:** $s_{n-1} = \sqrt{s_{n-1}^2}$; it shares the same physical unit as the original data / sample mean / IQR.
3. **Sample Range:** $\boxed{\text{Range} = x_{\max} - x_{\min}}$; useful in statistical quality control.
4. **Interquartile Range (IQR):** $\boxed{\text{IQR} = Q_3 - Q_1}$, the range of the middle 50% of the data; some outlier problems can be eliminated by using IQR.

Remarks: All of them cannot be negative, and they are zero if and only if all data points are identical (no spread). The larger they are, the more spread out (dispersed) the data. *Example (pH meter):* data $7.07, 7.00, 7.10, 6.97, 7.00, 7.03, 7.01, 7.01, 6.98, 7.08$ give $\overline{x} = 7.0250$, $s_{n-1}^2 = \dfrac{(7.07 - 7.025)^2 + (7.00 - 7.025)^2 + \cdots + (7.08 - 7.025)^2}{9} = 0.001939$, so $s_{n-1} = \sqrt{0.001939} = 0.0440$ with $n - 1 = 9$ degrees of freedom.

R corner (numerical summaries): R has built-in functions for every summary above. Note that `var()` and `sd()` use the $n-1$ denominator, matching this course.

```
x <- c(7.07, 7.00, 7.10, 6.97, 7.00, 7.03,
```

```
       7.01, 7.01, 6.98, 7.08)
mean(x)             # sample mean:   7.025
median(x)           # sample median
var(x)              # sample variance s^2 (n-1 denominator)
sd(x)               # sample standard deviation
range(x)            # min and max
max(x) - min(x)     # sample range
quantile(x, 0.25)   # Q1; 0.5 gives median, 0.75 gives Q3
IQR(x)              # interquartile range Q3 - Q1
summary(x)          # min, Q1, median, mean, Q3, max at once
```

Remark: R's `quantile()` offers 9 different quartile conventions (`type = 1, …, 9`, see `?quantile`); its default may differ slightly from the hand calculation rule above, so do not panic when the answers are not identical.

1.3 Tabular and Graphical Presentations

It is best to use graphical summaries to communicate information; the nature of the data and the goal of visualization determine which method to choose.

1.3.1 Presenting Quantitative Data

- **Line Chart:** Ideal for time-series data (e.g., stock prices, daily confirmed COVID-19 cases) since it visualizes trends clearly.
- **Frequency Table & Histogram:** A frequency table arranges data into numerically ordered, non-overlapping *class intervals* to condense data into a more useful form. *Procedure:* (1) determine the number of classes (normally 5–20); (2) obtain a suitable class width by dividing the range of the data by the number of classes; (3) establish the boundaries of each class to avoid overlapping; (4) construct a table with columns of class interval, class midpoint and class frequency; then build the *relative frequency distribution*. A **histogram** is the graphical version: class frequencies (or relative frequencies) on the y-axis, class intervals on the x-axis.
- **Boxplot:** see below.
- **Scatter Plot:** Depicts data that come as *pairs* (x_i, y_i) and visualizes the relationship between two quantitative variables (e.g., years of education vs. salary; cotton percentage vs. tensile strength).

R corner (line chart): `plot()` with `type = "b"` (both points and lines) draws a line chart for time-series data; `read.table()` imports external data.

```
z <- read.table(file.choose(), sep = "\t", header = TRUE)
plot(z$New_Daily_confirmed, type = "b", col = "blue",
     xlab = "Date index",
     ylab = "Number of confirmed cases (daily)",
     main = "COVID-19, Jan 2023, HKG")
```

R corner (histogram): for the 40 car-battery lives, `hist()` with `breaks = seq(...)` controls the class intervals; set `freq = FALSE` for a density histogram (total area = 1).

```
zz <- c(2.2, 4.1, 3.5, 4.5, 3.2, 3.7, 3.0, 2.6, 3.4, 1.6,
        3.1, 3.3, 3.8, 3.1, 4.7, 3.7, 2.5, 4.3, 3.4, 3.6,
        2.9, 3.3, 3.9, 3.1, 3.3, 3.1, 3.7, 4.4, 3.2, 4.1,
        1.9, 3.4, 4.7, 3.8, 3.2, 2.6, 3.9, 3.0, 4.2, 3.5)
hist(zz, breaks = seq(1.45, 4.95, by = 0.5),
```

```
     main = "Frequency histogram",
     xlab = "Battery life (years)", ylab = "Frequency",
     col = "light green")
hist(zz, breaks = seq(1.45, 4.95, by = 0.5), freq = FALSE,
     main = "Density histogram",
     xlab = "Battery life (years)", ylab = "Density",
     col = "light blue")
```

The `lattice` package offers an alternative: after `library(lattice)`, use `histogram(zz, breaks = seq(1.45, 4.95, by = 0.5))`.

R corner (scatter plot): give `plot()` the two paired variables; `pch` and `cex` control the point symbol and size, `rgb()` sets a translucent color.

```
df <- data.frame(CottonPercentage = rep(c(15, 20, 25, 30), each = 5),
                 TensileStrength = c(7, 7, 9, 8, 10, 19, 20, 21, 20, 22,
                                     21, 21, 17, 19, 20, 8, 7, 8, 9, \
   10))
plot(df$CottonPercentage, df$TensileStrength,
     xlab = "Cotton Percentage", ylab = "Tensile Strength",
     col = rgb(0, 0, 1, alpha = 0.3), pch = 16, cex = 2)
```

1.3.2 Skewness, Modality and Percentiles

- **Symmetric Distribution:** Mean $\approx$ Median.
- **Right-Skewed / Positively Skewed:** Long tail extends to the right; Mean $>$ Median (e.g., household income: mean $\approx$ \$67,000 vs. median $\approx$ \$50,000).
- **Left-Skewed / Negatively Skewed:** Long tail extends to the left; Mean $<$ Median.
- **Mean vs. Median:** the mean summarizes the center well if the distribution is symmetric, unimodal and has no outliers; otherwise the median is a better alternative.
- **Modality:** Unimodal, Bimodal, Multimodal, and Uniform shapes.
- **Percentiles / Quantiles:** The 90th percentile (0.90 quantile) of incomes being \$135,000 means 90% of households report an income of \$135,000 or less and 10% report more. The 75th percentile (0.75 quantile) is the **3rd quartile** Q_3; the 50th percentile is the **median** Q_2; the 25th percentile is the **1st quartile** Q_1.

1.3.3 Boxplot and Outlier Detection Rule

A boxplot (box-and-whisker plot) is constructed from the **Five-Number Summary**: the minimum, lower quartile (Q_1), median (Q_2), upper quartile (Q_3), and maximum. Quartiles split the ordered data into 4 segments with an equal number of data per segment: 25% of the data lie below Q_1 and 25% lie above Q_3. In practice Q_2 measures central tendency and $Q_3 - Q_1$ (the IQR) measures dispersion.

- **Quartile Calculation Rule:** Let n be the total number of data and $p \in \{0.25, 0.5, 0.75\}$. If $np + 0.5$ is an integer, say m, then the m-th smallest data point is the corresponding quartile. If $np + 0.5$ is NOT an integer, let $m = \lfloor np + 0.5 \rfloor$ (the greatest integer just less than it); the quartile is the average of the m-th and $(m+1)$-th smallest data points.
- **1.5 $\times$ IQR Outlier Rule:** An observation x is flagged as an outlier if $\boxed{x \le Q_1 - 1.5 \times \text{IQR} \quad \text{or} \quad x \ge Q_3 + 1.5 \times \text{IQR}}$. For normal data, this occurs with probability ≈ 0.007.

- **Side-by-side Boxplot:** used to compare grouped data across different groups.

R corner (boxplot): `boxplot(x, horizontal = TRUE)` draws a horizontal boxplot and marks outlier points beyond the whiskers automatically. Try the two data sets from the questions — the second has two obvious outliers $(57, 89)$.

```
x1 <- c(12, 13, 14, 15, 16, 17, 18, 18, 19, 20, 20, 21,
        22, 22, 23, 24, 25, 26, 26, 27, 28, 29, 29, 30,
        31, 32, 33, 34, 35, 42, 47)
boxplot(x1, horizontal = TRUE)
x2 <- c(3, 4, 12, 13, 14, 15, 16, 17, 18, 18, 19, 20, 20, 57, 89)
boxplot(x2, horizontal = TRUE)
```

For grouped data, build a data frame and use the formula interface `y ~ group` to obtain side-by-side boxplots:

```
df <- data.frame(team = rep(c('A', 'B', 'C'), each = 8),
                 points = c(5, 5, 6, 6, 8, 9, 13, 15,
                            11, 11, 12, 14, 15, 19, 22, 24,
                            19, 23, 23, 23, 24, 26, 29, 33))
head(df, 10)   # preview the first 10 rows
boxplot(df$points ~ df$team, col = 'steelblue',
        main = 'Points by Team', xlab = 'Team', ylab = 'Points')
```

1.3.4 Presenting Categorical Data

- **Frequency Table:** counts and percentages per category (with cumulative counts/percent for ordinal-level variables).
- **Pie Chart:** Displays proportions/percentages of categorical groups as sectors.
- **Bar Chart:** Uses rectangular bars to visually compare frequencies across distinct categories.

R corner (pie chart and bar chart): organize the categories and counts in a data frame, then call `pie()` or `barplot()` (use `table()` on raw categorical data to obtain the counts first).

```
z <- data.frame(Campus = c("University Park", "Commonwealth Campuses",
                           "PA College of Technology", "World Campus"),
                Count = c(40639, 27100, 4981, 8360))
pie(z$Count, labels = z$Campus,
    col = c("purple", "violetred1", "green3", "cornsilk"))
barplot(z$Count, names.arg = z$Campus,
        xlab = "Campus", ylab = "Count")
```

1.4 Sampling Methods and Biases

1.4.1 Pitfalls of Sampling (Sampling Bias)

Sampling correctly is very important; flawed techniques distort conclusions (e.g., **Survivorship Bias**):

- **Selection Bias:** A sample of convenience makes it more likely to sample certain subjects than others.
- **Non-Response Bias:** Occurs when sampled individuals fail or refuse to respond (e.g., parents busy with children and dinner at 6 pm).

- **Voluntary Response Bias:** Those with very bad or very good experiences are disproportionately represented (e.g., online business reviews).

1.4.2 Classifications of Sampling Techniques

1. **Probability Sampling:** Involves random selection, minimizing the risk of selection bias and allowing strong statistical inference about the whole group.
   - Simple Random Sampling
   - Systematic Sampling
   - Stratified Sampling
   - Cluster Sampling
2. **Non-Probability Sampling:** Non-random selection based on convenience or other criteria, allowing easy data collection.
   - Convenience Sampling
   - Voluntary Response Sampling
   - Purposive Sampling
   - Snowball Sampling

2 Probability

2.1 Sample Spaces and Events

2.1.1 Basic Concepts

- **Random Experiment:** A process whose outcome cannot be predicted with certainty prior to observation (e.g., rolling a die, tossing a coin, observing particle life).
- **Question of Chance:** Expresses the likelihood of specific outcomes (e.g., "How likely is it that the result is even?").
- **Probability Assignment:** A quantitative measure assigning a number in $[0, 1]$ to an event; a larger probability indicates a higher likelihood.
- **Deterministic Experiments:** modeled as special cases: if an outcome is guaranteed, its event has probability 1 (e.g., $\mathbb{P}(\text{result is } 4) = 1$ when the die always yields 4).

2.1.2 Remark: Probability Zero and One

A nonempty event can have probability zero, and a probability-one event need not contain every outcome. *Example:* pick a real number uniformly from $S = [0, 1]$: the singleton $\{0\}$ has measure zero so $\mathbb{P}(\{0\}) = 0$ even though $0 \in S$; the complement $(0, 1]$ satisfies $\mathbb{P}((0, 1]) = 1 - \mathbb{P}(\{0\}) = 1$ yet does not contain 0.

2.1.3 Fundamental Definitions

- **Sample Space (S):** the set of all possible outcomes of a random experiment. E.g., a die: $S = \{1, 2, 3, 4, 5, 6\}$; a coin: $S = \{H, T\}$.
- **Sample Point (s):** an individual outcome/element/member of S ($s \in S$).
- **Event (A):** any subset of S ($A \subseteq S$), collecting all sample points satisfying a condition. E.g., for $S = \{1, \ldots, 6\}$, "result is even" is $A = \{2, 4, 6\}$.

- **Occurrence:** A occurs iff the observed outcome s belongs to A ($s \in A$). One observed outcome can make multiple events occur simultaneously (4 is even and greater than 3).
- **Elements vs. Subsets:** $3 \in D$ means 3 is an element of $D = \{3, 6\}$; $\{3\} \subseteq D$ means the singleton is a subset; $\{3\} \notin D$ because D's elements are numbers, not sets.

2.1.4 Cardinality and Special Events

Let $|A|$ denote the cardinality (number of outcomes) of a finite event A.

- **Impossible Event ($\emptyset$):** the empty set, $|A| = 0$, $\mathbb{P}(\emptyset) = 0$. (Do not confuse $\emptyset$ with $\{0\}$, which has one element.)
- **Singleton Event:** exactly one outcome, $|A| = 1$ (e.g., $A = \{3\}$).
- **Certain Event (S):** the full sample space, $|S| = N$, $\mathbb{P}(S) = 1$.

2.1.5 Multi-Step Experiments and Sample Space Modeling

When an experiment has sequential stages, outcomes must capture the complete path. *Example:* (1) toss a coin; (2) if H, toss again; if T, roll a die. Then $S = \{HH, HT, T1, T2, T3, T4, T5, T6\}$ with $\mathbb{P}(\{HH\}) = \mathbb{P}(\{HT\}) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$ and $\mathbb{P}(\{Tk\}) = \frac{1}{2} \times \frac{1}{6} = \frac{1}{12}$ for $k = 1, \ldots, 6$; total: $2\left(\frac{1}{4}\right) + 6\left(\frac{1}{12}\right) = 1$.

2.1.6 Describing Sets and Continuous Sample Spaces

When outcomes are numerous or infinite, use set-builder notation $A = \{s \in S : s \text{ satisfies condition } P\}$. *Example (component lifetime t in hours):* $S = \{t : t \geq 0\} = [0, \infty)$; failure before 5 hours: $E = [0, 5)$; lifetime at least 5 hours: $F = [5, \infty)$; lifetime exactly 3 hours: $G = \{3\}$.

2.1.7 Cartesian Product for Repeated Experiments

When an experiment is repeated (e.g., a die rolled twice), ordered pairs preserve sequence and repetition: $S \times S = \{(s_1, s_2) : s_1 \in S, s_2 \in S\}$ with $|S \times S| = 36$; for $A = \{2, 4, 6\}$, $A \times A = \{(s_1, s_2) : s_1, s_2 \in A\}$ with $|A \times A| = 9$; under equiprobable outcomes $\mathbb{P}(A \times A) = \frac{9}{36} = \frac{1}{4}$.

2.1.8 Set Operations

Given events $A, B \subseteq S$:

1. **Union ($A \cup B$):** A or B occurs (inclusive or): $A \cup B = \{s \in S : s \in A \text{ or } s \in B\}$.
2. **Intersection ($A \cap B$):** both occur: $A \cap B = \{s \in S : s \in A \text{ and } s \in B\}$.
3. **Complement (A^c):** A does not occur: $A^c = \{s \in S : s \notin A\}$; properties: $S^c = \emptyset$, $\emptyset^c = S$, $(A^c)^c = A$, $A \cap A^c = \emptyset$, $A \cup A^c = S$.

4. **Set Difference ($A - B$ or $A \setminus B$):** A occurs but B does not: $A - B = \{s \in S : s \in A \text{ and } s \notin B\} = A \cap B^c$.

2.1.9 Relations Between Events

- **Containment ($A \subseteq B$):** every outcome of A is in B; if A occurs, B must occur. Note $A \subseteq B \implies A \cup B = B$, $A \cap B = A$. Proper subset: $A \subsetneq B$ means $A \subseteq B$ and $A \neq B$.
- **Mutual Exclusivity (Disjointness):** events cannot occur simultaneously: $A \cap B = \emptyset$ (at most one occurs). Pairwise disjoint collection: $A_i \cap A_j = \emptyset$ for all $i \neq j$.
- **Exhaustiveness:** the events collectively cover S: $\bigcup_{i=1}^{n} A_i = S$ (at least one occurs).
- **Partition:** a collection both pairwise disjoint and exhaustive.

2.1.10 Set Identities and Laws

Commutative: $A \cup B = B \cup A$, $A \cap B = B \cap A$. **Associative:** $(A \cup B) \cup C = A \cup (B \cup C)$, $(A \cap B) \cap C = A \cap (B \cap C)$. **Distributive:** $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$, $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$. **De Morgan's Laws:** $(A \cup B)^c = A^c \cap B^c$ ("neither A nor B occurs"); $(A \cap B)^c = A^c \cup B^c$ ("not both A and B occur").

2.2 Probability (Kolmogorov's Axiom System)

2.2.1 Axiomatic Definition

In 1933 Kolmogorov published the axiomatic structure of probability theory.

Definition: A probability on the sample space S is an assignment of a value $\mathbb{P}(E)$ to each event E via a function $\mathbb{P}(\cdot)$ such that:

1. **Non-negativity:** $\mathbb{P}(E) \geq 0$ for any event E in S.
2. **Normalization:** $\mathbb{P}(S) = 1$.
3. **Countable Additivity:** for any sequence of mutually exclusive events $E_1, E_2, \ldots$:

$$\boxed{\mathbb{P}\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(E_i)}.$$

Remark (Countable): A set A is countable if it is finite or its elements can be enumerated as a sequence $a_1, a_2, a_3, \ldots$, i.e., $A = \bigcup_{k=1}^{\infty} \{a_k\}$; equivalently, a countably infinite set is in one-to-one correspondence with $\mathbb{N} = \{1, 2, 3, \ldots\}$. $\mathbb{N}$ is countable; $\mathbb{R}$ and $[0, 1]$ are not. Interpretation: $E \mapsto \mathbb{P}(E)$ is a special "function" and probability is a "measure".

2.2.2 Consequences and Derived Properties

1. Probability of the empty set: $\boxed{\mathbb{P}(\emptyset) = 0}$.

Proof: Since $\emptyset = \emptyset \cup \emptyset \cup \cdots$, countable additivity gives $\mathbb{P}(\emptyset) = \mathbb{P}(\emptyset) + \mathbb{P}(\emptyset) + \cdots$; as $\mathbb{P}(\emptyset)$ is a real number, it must be 0.

2. Finite Additivity: $\boxed{\mathbb{P}\left(\bigcup_{i=1}^{m} A_i\right) = \sum_{i=1}^{m} \mathbb{P}(A_i)}$ for mutually exclusive $A_1, \ldots, A_m$.

Proof: append empty sets: $\bigcup_{i=1}^{m} A_i = A_1 \cup \cdots \cup A_m \cup \emptyset \cup \emptyset \cup \cdots$, so by countable additivity $\mathbb{P}\left(\bigcup_{i=1}^{m} A_i\right) = \sum_{i=1}^{m} \mathbb{P}(A_i) + \mathbb{P}(\emptyset) + \cdots = \sum_{i=1}^{m} \mathbb{P}(A_i)$.

3. Monotonicity: $\boxed{A \subseteq B \implies \mathbb{P}(A) \leq \mathbb{P}(B)}$.

Proof: since $A \subseteq B$, $B = A \cup (B - A)$ with A and $B - A$ disjoint; finite additivity gives $\mathbb{P}(B) - \mathbb{P}(A) = \mathbb{P}(B - A) \geq 0$ by non-negativity. *Remark:* $\mathbb{P}(A) = \mathbb{P}(B)$ if $A = B$.

4. Numeric Bounds: $\boxed{0 \leq \mathbb{P}(A) \leq 1}$ (since $\emptyset \subseteq A \subseteq S$).

5. Complement Rule: $\boxed{\mathbb{P}(A^c) = 1 - \mathbb{P}(A)}$.

Proof: $A \cup A^c = S$ and $A \cap A^c = \emptyset$, so finite additivity gives $\mathbb{P}(A) + \mathbb{P}(A^c) = \mathbb{P}(S) = 1$.

6. Addition Rule (2 events): $\boxed{\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)}$.

Proof: decompose $A \cup B = A \cup (B - A)$ and $B = (A \cap B) \cup (B - A)$ into disjoint parts; then $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B - A)$ and $\mathbb{P}(B - A) = \mathbb{P}(B) - \mathbb{P}(A \cap B)$; substituting yields the rule.

Card examples: $\mathbb{P}(\text{Jack or 3}) = \frac{4}{52} + \frac{4}{52} \approx 0.154$ (disjoint); $\mathbb{P}(\text{Jack or red}) = \frac{4}{52} + \frac{26}{52} - \frac{2}{52} \approx 0.538$ (non-disjoint).

Addition Rule (3 events): $\mathbb{P}(A \cup B \cup C) = \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) - \mathbb{P}(A \cap B) - \mathbb{P}(A \cap C) - \mathbb{P}(B \cap C) + \mathbb{P}(A \cap B \cap C)$, obtained by treating $(A \cup B)$ as one event and applying the two-event rule with $(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$.

Inclusion–Exclusion Principle: for any events $A_1, \ldots, A_n$: $\mathbb{P}(A_1 \cup \cdots \cup A_n) = \sum_{i=1}^{n} \mathbb{P}(A_i) - \sum_{1 \leq i < j \leq n} \mathbb{P}(A_i \cap A_j) + \sum_{1 \leq i < j < k \leq n} \mathbb{P}(A_i \cap A_j \cap A_k) - \cdots + (-1)^{n-1} \mathbb{P}(A_1 \cap \cdots \cap A_n)$. If $A_1, \ldots, A_n$ are mutually exclusive, all intersection terms vanish: $\mathbb{P}(A_1 \cup \cdots \cup A_n) = \mathbb{P}(A_1) + \cdots + \mathbb{P}(A_n)$.

Example 1: if $\mathbb{P}(A) = \mathbb{P}(B) = \mathbb{P}(C) = \frac{1}{4}$, $\mathbb{P}(A \cap B) = \mathbb{P}(B \cap C) = 0$, $\mathbb{P}(A \cap C) = \frac{1}{8}$, then $\mathbb{P}(\text{at least one}) = \frac{3}{4} - \frac{1}{8} + 0 = \frac{5}{8}$ (note $A \cap B \cap C \subseteq A \cap B$ forces $\mathbb{P}(A \cap B \cap C) = 0$). *Example 2:* $\mathbb{P}(\text{offer from A}) = 0.8$, $\mathbb{P}(\text{B}) = 0.6$, $\mathbb{P}(\text{both}) = 0.5$, so $\mathbb{P}(\text{at least one}) = 0.8 + 0.6 - 0.5 = 0.9$.

Example (weather): statuses are only Sunny, Cloudy, Rainy, with $\mathbb{P}(\text{Sunny}) = 2\,\mathbb{P}(\text{Cloudy})$ and $\mathbb{P}(\text{Rainy}) = \frac{1}{2}\,\mathbb{P}(\text{Cloudy})$. Normalization $\left(2 + 1 + \frac{1}{2}\right)\mathbb{P}(\text{Cloudy}) = 1$ yields $\mathbb{P}(\text{Sunny}) = \frac{4}{7}$, $\mathbb{P}(\text{Cloudy}) = \frac{2}{7}$, $\mathbb{P}(\text{Rainy}) = \frac{1}{7}$.

2.3 Counting Techniques

2.3.1 Finite Equally Likely Outcomes

Outcomes are *equally likely* if they have equal chances to occur.

Theorem: for $S = \{s_1, \ldots, s_N\}$ with N equally likely outcomes and any event $E \subseteq S$: $\boxed{\mathbb{P}(E) = \dfrac{\text{number of outcomes in } E}{N}}$.

Proof: if $E = \{s_{i_1}, \ldots, s_{i_k}\}$, then $\mathbb{P}(E) = \mathbb{P}(s_{i_1}) + \cdots + \mathbb{P}(s_{i_k}) = \dfrac{1}{N} + \cdots + \dfrac{1}{N} = \dfrac{k}{N}$. **Attention:** the formula holds only if all outcomes are equally likely! If a fair coin is thrown twice and only the *number of heads* is recorded, $S = \{0, 1, 2\}$ is NOT equally likely, and $\mathbb{P}(\text{at least one head}) \neq \dfrac{2}{3}$; recording the sequence gives $S = \{HH, HT, TH, TT\}$, $E = \{HH, HT, TH\}$ and $\mathbb{P}(E) = \dfrac{3}{4}$. *Example:* two dice thrown once: $|S| = 36$; $\mathbb{P}(\text{sum} = 5) = \dfrac{4}{36} = \dfrac{1}{9}$ without listing. *Counterintuitive (Simpson's Paradox):* choosing among boxes with $(5R, 6G)$, $(3R, 4G)$, $(6R, 3G)$, $(9R, 5G)$ balls can defy intuition when aggregating.

R corner: `sample()` mimics a random experiment, and repeating it many times lets us estimate a probability by relative frequency — the estimate approaches $\dfrac{1}{9}$ as M grows (the frequentist view of probability).

```
sample(x = 1:6, size = 2, replace = TRUE)   # one throw of 2 dice
# Repeat the experiment M times and count how often the sum is 5:
M <- 1000
dicesum <- numeric(M)
for (i in 1:M) {
  dicesum[i] <- sum(sample(x = 1:6, size = 2, replace = TRUE))
}
sum(dicesum == 5)          # number of times sum = 5
sum(dicesum == 5) / M      # "estimate" of P(sum = 5), close to 1/9
```

2.3.2 Addition Principle

If there are m_1 outcomes from the 1st way, m_2 from the 2nd way, …, m_n from the n-th way (n different *ways*, mutually exclusive), the total number of outcomes is $\boxed{N = m_1 + m_2 + \cdots + m_n}$.

2.3.3 Multiplication Principle

If a procedure consists of n *steps* (in sequence) with m_1 outcomes in the 1st step, m_2 in the 2nd, …, m_n in the n-th, the total number of outcomes is $\boxed{N = m_1 m_2 \cdots m_n}$. *Example (computer assembly):* chips from 2 brands, hard drives from 4, memory from 3, accessory bundles from 5: $2 \times 4 \times 3 \times 5 = 120$ ways. *52-card deck:* 13 face values $\times$ 4 suits (Hearts, Diamonds, Spades, Clubs).

2.3.4 Permutation

Definition: a permutation is an *ordered* arrangement of all or part of a set of objects. For n distinct objects, arranging all of them takes n steps with $n, n-1, \ldots, 1$ options: $n \times (n-1) \times \cdots \times 2 \times 1$ ways.

Definition (factorial): for a non-negative integer n, $n! = n(n-1)\cdots(2)(1)$, with special case $0! = 1$.

Theorem: the number of permutations of n distinct objects is $n!$.

Theorem: the number of permutations of n distinct objects taken r at a time is $\boxed{P_r^n = \dfrac{n!}{(n-r)!}}$ (fill r slots: $n, n-1, \ldots, n-r+1$ options).

Example (officers): a president and a treasurer are chosen from a club of 50 people: (a) no restrictions: $50 \times 49 = 2450$; (b) A serves only if president: $49 + 49 \times 48 = 2401$; (c) B and C serve together or not at all: $2 + 48 \times 47 = 2258$; (d) D and E will not serve together: $2450 - 2 = 2448$.

Example (Birthday Problem): in a room of n people ($n \leq 365$), with each day equally likely as a birthday, let $A =$ "at least two share a birthday". Since $\mathbb{P}(A^c)$ is easier: $|S| = 365^n$ and A^c happens in $365 \times 364 \times \cdots \times (365-n+1)$ ways, so $\mathbb{P}(A) = 1 - \dfrac{365 \times 364 \times \cdots \times (365-n+1)}{365^n}$. Values: $n = 20 : 0.41$; $25 : 0.57$; $30 : 0.71$; $40 : 0.89$; $50 : 0.97$; $55 : 0.99$; $100 : 0.9999997$.

Theorem (Circular Permutation): the number of permutations of n objects arranged in a circle is $\boxed{(n-1)!}$: two circular permutations are not different unless corresponding objects are preceded or followed by a different object clockwise. *Example:* 4 people playing bridge around a table: $(4-1)! = 3! = 6$ distinct arrangements (everyone moving one position clockwise is the same arrangement).

2.3.5 Combination

In many problems we select r objects from n *without regard to order*: these are combinations.

Theorem: the number of combinations of n distinct objects taken r at a time is $\boxed{\binom{n}{r} = \dfrac{n!}{(n-r)!\,r!}}$. *Remark:* by the multiplication principle, (number of ordered samples) = (number of unordered samples) $\times$ (number of ways to order each sample), i.e., $\dfrac{n!}{(n-r)!} = \binom{n}{r} \cdot r!$. *Example:* choosing 5 cartridges: 3 arcade from 10 and 2 sports from 5: $\binom{10}{3} \cdot \binom{5}{2} = 120 \times 10 = 1200$ ways.

2.3.6 Partition (Multinomial Coefficient)

Theorem: the number of ways that n distinct objects can be grouped into k classes with n_i objects in the i-th class ($i = 1, \ldots, k$, $\sum_{i=1}^{k} n_i = n$) is $\boxed{\binom{n}{n_1, n_2, \ldots, n_k} = \frac{n!}{n_1!\, n_2! \cdots n_k!}}$. (Special case $k = 2$ gives $\binom{n}{n_1}$.) *Example:* assign 7 graduate students to 1 triple and 2 double hotel rooms: $\binom{7}{3, 2, 2} = \frac{7!}{3!\, 2!\, 2!} = 210$.

2.3.7 Drawing Order Does Not Matter

Example: a bag contains M red and N green balls; draw one ball each time; let $E =$ "a red ball at the k-th draw" ($1 \le k \le M + N$). **Solution I (distinct balls, permutation):** all outcomes: $(M + N)!$; outcomes in E: fix a red ball at spot k (M choices) and arrange the other $M + N - 1$ balls: $M \cdot (M + N - 1)!$, so $\mathbb{P}(E) = \frac{M(M+N-1)!}{(M+N)!} = \frac{M}{M+N}$. **Solution II (indistinguishable balls, combination):** the k-th spot takes any red ball, choose $M - 1$ red spots from the remaining $M + N - 1$: $\mathbb{P}(E) = \frac{\binom{M+N-1}{M-1}}{\binom{M+N}{M}} = \frac{M}{M+N}$. **Remarks:** Solution I uses permutation, Solution II combination; the probability of red at the k-th draw does NOT depend on k. *Similar example (lucky draw):* a class of n students draws for m concert tickets ($m < n$): whether one gets a ticket has nothing to do with the drawing order; the probability is the constant $\frac{m}{n}$.

2.4 Conditional Probability

2.4.1 Motivation and Definition

Lucky draw revisited: n students draw for m tickets ($m < n$). If you are the k-th person to draw but it is known that the first $k - 1$ people drew no tickets, your probability is no longer $\frac{m}{n}$ but $\frac{m}{n-k+1}$. Sometimes the information that an event B has occurred influences the probability of A: we "update" $\mathbb{P}(A)$ with additional information from B, written $\mathbb{P}(A \mid B)$.

Die example: toss a fair die once: $\mathbb{P}(\text{odd}) = \frac{1}{2}$, but $\mathbb{P}(\text{odd} \mid \text{number} > 1) = \frac{2}{5}$ since the sample space is reduced to $\{2, 3, 4, 5, 6\}$.

Definition: let A, B be events with $\mathbb{P}(B) > 0$; the conditional probability of A given B is $\boxed{\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}}$. **Remarks:** conditional probability is meaningless when $\mathbb{P}(B) = 0$; $\mathbb{P}(A \mid B) \ge \mathbb{P}(A \cap B)$. In the die example, $\mathbb{P}(A \mid B) = \frac{2/6}{5/6} = \frac{2}{5}$ with $A =$ odd, $B =$ greater than 1.

2.4.2 Multiplicative Rule (Product Rule)

Rearranging the definition: $\boxed{\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)\,\mathbb{P}(B)}$ (provided $\mathbb{P}(B) > 0$); symmetrically $\mathbb{P}(A \cap B) = \mathbb{P}(B \mid A)\,\mathbb{P}(A)$ (provided $\mathbb{P}(A) > 0$). *Example (two bags):* bag 1 has 2 black and 2 white balls; bag 2 has 3 black and 3 white. One ball is moved from bag 1 to bag 2, then one ball is drawn from bag 2. With A_i = "white from the i-th pick": $\mathbb{P}(A_1) = \frac{1}{2}$, $\mathbb{P}(A_2 \mid A_1) = \frac{4}{7}$, so $\mathbb{P}(A_1 \cap A_2) = \mathbb{P}(A_2 \mid A_1)\mathbb{P}(A_1) = \frac{4}{7} \cdot \frac{1}{2} = \frac{2}{7}$.

Generalization (n events): if $\mathbb{P}(A_1 \cap A_2 \cap \cdots \cap A_{n-1}) > 0$, then $\mathbb{P}(A_1 \cap A_2 \cap \cdots \cap A_n) = \mathbb{P}(A_n \mid A_1 \cap \cdots \cap A_{n-1})\,\mathbb{P}(A_{n-1} \mid A_1 \cap \cdots \cap A_{n-2}) \cdots \mathbb{P}(A_2 \mid A_1)\,\mathbb{P}(A_1)$. For instance, $\mathbb{P}(A \cap B \cap C) = \mathbb{P}(C \mid A \cap B)\,\mathbb{P}(B \mid A)\,\mathbb{P}(A)$. *Example (fuses):* a box of 20 fuses contains 5 defectives; three fuses are drawn in succession without replacement: $\mathbb{P}(A) = \frac{5}{20}$, $\mathbb{P}(B \mid A) = \frac{4}{19}$, $\mathbb{P}(C \mid A \cap B) = \frac{3}{18}$, so $\mathbb{P}(A \cap B \cap C) = \frac{3}{18} \times \frac{4}{19} \times \frac{5}{20} = \frac{1}{114}$.

2.4.3 Properties of Conditional Probability

The **additive rule** computes "at least one": $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$; the **multiplicative rule** computes "both": $\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)\mathbb{P}(B)$; both generalize. Some properties: $\mathbb{P}(A \mid B) = 1 - \mathbb{P}(A^c \mid B)$; $\mathbb{P}((A \cup B) \mid D) = \mathbb{P}(A \mid D) + \mathbb{P}(B \mid D) - \mathbb{P}((A \cap B) \mid D)$; $\mathbb{P}((A \cap B) \mid D) = \mathbb{P}(A \mid (B \cap D))\,\mathbb{P}(B \mid D)$.

2.4.4 Law of Total Probability

To compute $\mathbb{P}(A)$ in a complicated scenario, split by a condition B: since $(A \cap B)$ and $(A \cap B^c)$ are disjoint and $A = (A \cap B) \cup (A \cap B^c)$, we get $\mathbb{P}(A) = \mathbb{P}(A \cap B) + \mathbb{P}(A \cap B^c)$, hence for $0 < \mathbb{P}(B) < 1$: $\boxed{\mathbb{P}(A) = \mathbb{P}(A \mid B)\,\mathbb{P}(B) + \mathbb{P}(A \mid B^c)\,\mathbb{P}(B^c)}$.

Partition of the sample space: events $B_1, \ldots, B_n$ form a partition of S if (1) they are mutually exclusive ($B_i \cap B_j = \emptyset$ for $i \neq j$) and (2) exhaustive ($B_1 \cup \cdots \cup B_n = S$). **Theorem (general version):** if $B_1, \ldots, B_n$ are mutually exclusive and exhaustive with $0 < \mathbb{P}(B_i) < 1$, then for any event A: $\boxed{\mathbb{P}(A) = \sum_{i=1}^{n} \mathbb{P}(A \mid B_i)\,\mathbb{P}(B_i)}$. (Tree diagrams organize such computations.)

Example (bag with replacement and addition): a bag has m red and n green balls. First pick: $\mathbb{P}(\text{green}) = \frac{n}{m+n}$. Record the color, put the ball back and add one ball of the same color; then pick again. With A = green at 2nd pick, B_1 = green at 1st, B_2 = red at 1st: $\mathbb{P}(A) = \frac{n}{m+n} \cdot \frac{n+1}{m+n+1} + \frac{m}{m+n} \cdot \frac{n}{m+n+1} = \frac{n}{m+n}$. (Question: what if adding c balls of the same color?)

Example (defective product): machines I, II, III make 30%, 45%, 25% of the products with defect rates 2%, 3%, 2%. With A = defective, B_i = made by machine i: $\mathbb{P}(A) = 0.3 \times 0.02 + 0.45 \times 0.03 + 0.25 \times 0.02 = 0.0245$.

2.4.5 Bayes' Theorem

Theorem (1st form): if $\mathbb{P}(A), \mathbb{P}(B) > 0$, then $\boxed{\mathbb{P}(B \mid A) = \dfrac{\mathbb{P}(B)\,\mathbb{P}(A \mid B)}{\mathbb{P}(A)}}$: it relates $\mathbb{P}(A \mid B)$ to its "inverse" $\mathbb{P}(B \mid A)$.

Theorem (2nd form): if $B_1, \ldots, B_n$ form a partition of S with $0 < \mathbb{P}(B_i) < 1$, then for any B_j: $\boxed{\mathbb{P}(B_j \mid A) = \dfrac{\mathbb{P}(A \mid B_j)\,\mathbb{P}(B_j)}{\sum_{i=1}^{n} \mathbb{P}(A \mid B_i)\,\mathbb{P}(B_i)}}$.

Defective product revisited: given the product is defective, $\mathbb{P}(\text{machine III}) = \dfrac{0.25 \times 0.02}{0.3 \times 0.02 + 0.45 \times 0.03 + 0.25 \times 0.02} = \dfrac{0.005}{0.0245} = \dfrac{10}{49}$.

Interpretation: the unconditional $\mathbb{P}(B_j)$ is the **prior probability** (no information about A); $\mathbb{P}(B_j \mid A)$ is the **posterior probability** (occurrence of A taken into account).

Example (diagnostic test): A = test shows positive for a virus, B = person infected. Given $\mathbb{P}(B) = 0.0004$, $\mathbb{P}(A \mid B) = 0.95$, $\mathbb{P}(A^c \mid B^c) = 0.90$: $\mathbb{P}(B \mid A) = \dfrac{0.95 \times 0.0004}{0.95 \times 0.0004 + 0.10 \times 0.9996} = 0.0038$ — a positive test barely raises the infection probability (how to interpret?).

Application (Warner's randomized response, 1960s): to estimate the prevalence p of a sensitive behavior (e.g., bullying), ask randomly one of two versions: (A_1) "you have bullied a classmate in the last two years?" or (A_2) "you have not bullied…" (T/F), chosen by drawing from a box with 2 red and 1 green balls (red $\to A_1$). The interviewer cannot tell which version was answered, so the respondent may answer truthfully. With B = "answers TRUE" and $q = \mathbb{P}(A_1)$: $\mathbb{P}(B) = q\,p + (1-q)(1-p)$; with $q = \dfrac{2}{3}$: $p = 3\,\mathbb{P}(B) - 1$, estimated by $\dfrac{3m}{n} - 1$ if m out of n answer TRUE; in general $\hat{p} = \dfrac{\frac{m}{n} - (1-q)}{2q-1}$.

Application (The Boy Who Cried Wolf, Bayesian update): A = boy lies, B = villagers trust him; $\mathbb{P}(A \mid B) = 0.1$, $\mathbb{P}(A \mid B^c) = 0.5$, $\mathbb{P}(B) = 0.8$. After one lie: $\mathbb{P}(B \mid A) = \dfrac{0.1 \times 0.8}{0.1 \times 0.8 + 0.5 \times 0.2} = 0.444$; after two lies (prior = 0.444): $\mathbb{P}(B \mid A) = \dfrac{0.1 \times 0.444}{0.1 \times 0.444 + 0.5 \times 0.556} = 0.138$. *Another application:* the Monty Hall problem (stick or switch?) can be resolved by evaluating $\mathbb{P}(B_1 \mid A)$ with Bayes' theorem, where B_i = car behind Door i and A = host opens Door III.

2.5 Independence

Events A and B are **independent** if $\mathbb{P}(A \cap B) = \mathbb{P}(A)\,\mathbb{P}(B)$; equivalently (for $\mathbb{P}(B) > 0$) $\mathbb{P}(A \mid B) = \mathbb{P}(A)$, i.e., knowing B does not change the probability of A. Random variables X, Y are **independent** if $\mathbb{P}(X \le x, Y \le y) = \mathbb{P}(X \le x)\,\mathbb{P}(Y \le y)$ for any $x, y \in \mathbb{R}$. Independence is used later: for independent r.v.'s, $\mathrm{Var}(X \pm Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$.

3 Random Variables

3.1 Random Variables (r.v.)

3.1.1 Motivation

Chapter 2 computes $\mathbb{P}(E)$ directly on S, e.g., tossing a fair coin twice: $S = \{HH, HT, TH, TT\}$, $E = \{HH, HT, TH\}$, $\mathbb{P}(E) = \frac{3}{4}$. But what if there are infinitely many outcomes, outcomes are not equally likely, or we want to apply calculus tools? Assigning *numbers* to outcomes (a "density" $f(x)$) opens the door to such tools. *Example:* toss a coin once, $S = \{H, T\}$, and define $X(s) = 0$ if $s = H$ and $X(s) = 1$ if $s = T$; then $X = 0 \iff s = H$ and $X = 1 \iff s = T$: X maps $S \to \mathbb{R}$ and is called a random variable.

3.1.2 Definition and Remarks

Definition: a random variable $X : S \to \mathbb{R}$ is a numerical valued function defined on a sample space: a number $X(s)$ is assigned to each outcome $s \in S$.

- R.v.'s are denoted by capital letters (X, Y, Z, W); their possible numerical values (*realizations*) by the corresponding lowercase letters (x, y, z, w).
- "Random" reminds us that outcomes (hence the values) cannot be predicted beforehand.
- New interpretation of DATA: data are the actual values (realizations) of the corresponding random variable.
- A r.v. X is a *function*, not a number; its value depends on the outcome.
- Events induced by X: $\{X = x\} = \{s \in S : X(s) = x\}$ and $\{X \leq x\} = \{s \in S : X(s) \leq x\}$. E.g., tossing a coin twice with $X =$ number of heads: $X = 1 \iff \{HT\}$ or $\{TH\}$ occurs.

3.1.3 Computing Probabilities

In general $\mathbb{P}(X = x) = \mathbb{P}(\{s \in S : X(s) = x\})$ where $x \in \mathbb{R}$. *Example (two flips, X = # heads):* $\mathbb{P}(X = 1) = \mathbb{P}(\{HT\} \cup \{TH\}) = \mathbb{P}(HT) + \mathbb{P}(TH) = \frac{1}{2}$; $\mathbb{P}(X \leq 1) = \mathbb{P}(\{TT, HT, TH\}) = \frac{3}{4}$, or via the complement $\{X \leq 1\} = S - \{X = 2\}$: $\mathbb{P}(X \leq 1) = 1 - \mathbb{P}(X = 2) = 1 - \frac{1}{4} = \frac{3}{4}$.

3.1.4 Types of Random Variables

The **range** of a r.v. is the collection of all possible values it can take on.

- **Discrete r.v.:** finite or countably infinite range. *Examples:* the number of defective items randomly selected from all items produced; the number of sales for a store; tossing a coin until a head turns up and defining $X =$ total number of tosses.
- **Continuous r.v.:** range is an interval of the real line (not rigorous, but enough for us). *Examples:* weight, length of an object; time until failure of a mechanical component; a city's daily average temperature.

3.2 Discrete Random Variables

3.2.1 Specification and Distribution

If a r.v. X takes only a finite or at most countably infinite number of values, X is a **discrete random variable**: possible values are isolated numbers. The range is denoted χ, e.g., $\chi = \{0, 1, 2\}$, $\chi = \{1, 2, \dots\}$, or $\chi = \{-1, 0.5, 2.7, 10.2\}$. To specify a discrete r.v. we need all possible values $x_1, x_2, \dots, x_k, \dots$ and their probabilities $p(x_k) = \mathbb{P}(X = x_k)$. A **(probability) distribution** of a r.v. is the collection of all possible values together with the probability of each.

Example (two dice): roll a pair of fair dice, X = their sum ($X(s_1) = X(1,2) = 3$, $X(s_2) = X(6,5) = X(s_3) = X(5,6) = 11$, …). With $p(x) = \mathbb{P}(X = x) = \frac{\#\text{ways 2 dice can sum to } x}{36}$:

x	2	3	4	5	6	7	8	9	10	11	12
$p(x)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

3.2.2 Probability Mass Function (pmf)

Definition: the pmf $p(x)$ of a discrete r.v. X gives the probability of occurrence for each possible value x of X (valid for all possible x). **Properties:** (i) $0 < p(x) \le 1$ for all $x \in \chi$ and $p(x) = 0$ for all $x \notin \chi$; (ii) $\sum_{\text{all } x \in \chi} p(x) = 1$. *Example (two flips, X = # heads, range $\{0,1,2\}$):* $p(0) = \frac{1}{4}$, $p(1) = \frac{1}{2}$, $p(2) = \frac{1}{4}$.

Probability of an event: for $A \subseteq \mathbb{R}$, $\boxed{\mathbb{P}(X \in A) = \sum_{x \in A} p(x)}$. With $\chi = \{1, 2, 5, 7\}$: for $A = [0, 5)$: $\mathbb{P}(X \in A) = p(1) + p(2)$; for $B = [0, 5]$: $p(1) + p(2) + p(5)$; for $C = \{1, 2\}$: $p(1) + p(2)$; for $D = \{2, 6\}$: $p(2)$.

3.2.3 Cumulative Distribution Function (cdf)

Alternatively a r.v. can be specified by its cdf.

Definition: the (cumulative) distribution function of a r.v. X is $\boxed{F(x) = \mathbb{P}(X \le x)}$ for all real x. **Remarks:** the cdf is defined for *all* random variables (including continuous ones); for a discrete r.v., $F(a) = \mathbb{P}(X \le a) = \sum_{x \le a} p(x)$ for all real a; the cdf is useful for intervals because for any $a < b$, $\mathbb{P}(a < X \le b) = F(b) - F(a)$. **Properties:** $F(x)$ is non-decreasing; $0 \le F(x) \le 1$ with $\lim_{x \to -\infty} F(x) = 0$ and $\lim_{x \to \infty} F(x) = 1$.

Example (pmf → cdf): given $p(1) = 0.3$, $p(2) = 0.2$, $p(3) = 0.5$: $F(x) = \begin{cases} 0, & x < 1 \\ 0.3, & 1 \le x < 2 \\ 0.5, & 2 \le x < 3 \\ 1, & x \ge 3 \end{cases}$ (for $x < 1$: $F(x) = \mathbb{P}(\emptyset) = 0$; for $1 \le x < 2$: $F(x) = p(1) = 0.3$; for $2 \le x < 3$: $p(1) + p(2) = 0.5$; for $x \ge 3$: $\mathbb{P}(S) = 1$).

Remarks: the cdf of a *discrete* r.v. is a *step function* with jumps of size $p(a)$ at possible values a (it only increases at jump points). If the range is $\{x_1, x_2, x_3, \dots\}$ with $x_1 < x_2 < x_3 < \cdots$, then $p(x_1) = F(x_1)$ and $p(x_j) = F(x_j) - F(x_{j-1})$ for $j = 2, 3, \dots$ (pmf $\leftrightarrow$ cdf). In the example: $\mathbb{P}(0.5 < X \le 2) = F(2) - F(0.5) = 0.5 - 0 = 0.5$; $\mathbb{P}(0.5 < X < 2) = \mathbb{P}(X = 1) = 0.3$.

3.2.4 Population Mean (Expectation)

Definition: if X is discrete with pmf $p(x)$ and range χ, the **population mean** (expectation, expected value) is $\boxed{\mathbb{E}(X) = \sum_{\text{all } x \in \chi} x\, p(x)}$, provided $\sum_{\text{all } x \in \chi} |x|\, p(x) < +\infty$ (i.e., the sum converges absolutely). **Remarks:** denoted μ or μ_X; a measure of central location of X; essentially a weighted average of all possible outcomes since $\sum_{\text{all } x} p(x) = 1$; it is a *number*, different from the sample mean (the mean of data).

Expectation of a function: for a real-valued function g, $\mathbb{E}[g(X)] = \sum_{\text{all } x \in \chi} g(x)\, p(x)$.

Example: if $\chi = \{-1, 0, 1, 2\}$ and $g(X) = X^2$, the distribution of $g(X)$ is $\{0, 1, 4\}$ with probabilities $p(0)$, $p(-1) + p(1)$, $p(2)$, so $\mathbb{E}(X^2) = 0 \cdot p(0) + 1 \cdot (p(-1) + p(1)) + 4 \cdot p(2) = (-1)^2 p(-1) + 0^2 p(0) + 1^2 p(1) + 2^2 p(2) = \sum_{\text{all } x} x^2 p(x)$.

Properties of expectation (when it exists; X, Y r.v.'s, a, b constants): $\mathbb{E}(b) = b$; $\mathbb{E}(aX) = a\,\mathbb{E}(X)$; $\mathbb{E}(aX + b) = a\,\mathbb{E}(X) + b$; $\mathbb{E}(X + Y) = \mathbb{E}(X) + \mathbb{E}(Y)$.

Example (car wash): the number of cars X between 4:00 and 5:00 P.M. on a sunny Friday has distribution $x = 4, 5, 6, 7, 8, 9$ with $p(x) = \frac{1}{12}, \frac{1}{12}, \frac{1}{4}, \frac{1}{4}, \frac{1}{6}, \frac{1}{6}$; the attendant earns $g(X) = 2X - 1$ dollars: $\mathbb{E}[g(X)] = \mathbb{E}(2X - 1) = \sum_{x=4}^{9} (2x - 1)\, p(x) = \$12.67$; alternatively $\mathbb{E}(2X - 1) = 2\,\mathbb{E}(X) - 1$.

Example (expected commission): a salesperson has a 70% chance of earning \$1000 at the first appointment and a 40% chance of earning \$1500 at the second (independent). The four possible totals $\$0, \$1000, \$1500, \$2500$ have probabilities $p(\$0) = (0.3)(0.6) = 0.18$, $p(\$1000) = (0.7)(0.6) = 0.42$, $p(\$1500) = (0.3)(0.4) = 0.12$, $p(\$2500) = (0.7)(0.4) = 0.28$; hence $\mathbb{E}(X) = 0(0.18) + 1000(0.42) + 1500(0.12) + 2500(0.28) = \$1300$. Shortcut: $\mathbb{E}(X_1 + X_2) = 0.7(1000) + 0.4(1500) = 700 + 600 = 1300$.

3.2.5 Population Variance

Definition: if X is discrete with pmf $p(x)$ and range χ, the **population variance** is $\boxed{\mathrm{Var}(X) = \sum_{\text{all } x \in \chi} (x - \mu)^2\, p(x)}$, if the sum exists. **Remarks:** denoted σ^2 or σ_X^2; the positive square root σ is the **population standard deviation (sd)**; both are common measures of spread/dispersion of X; they are *numbers*, different from the sample variance

(sd) of data; letting $g(X) = (X-\mu)^2$, we have $\mathrm{Var}(X) = \mathbb{E}[g(X)] = \mathbb{E}[(X-\mu)^2]$.

Shortcut formula: $\boxed{\mathrm{Var}(X) = \mathbb{E}(X^2) - [\mathbb{E}(X)]^2}$, i.e., $\sigma^2 = \mathbb{E}(X^2) - \mu^2$.

Proof: $\sigma^2 = \mathbb{E}[(X-\mu)^2] = \mathbb{E}(X^2 - 2\mu X + \mu^2) = \mathbb{E}(X^2) + \mathbb{E}(-2\mu X) + \mathbb{E}(\mu^2) = \mathbb{E}(X^2) - 2\mu\,\mathbb{E}(X) + \mu^2 = \mathbb{E}(X^2) - 2\mu^2 + \mu^2 = \mathbb{E}(X^2) - \mu^2$.

Properties of variance (when it exists; X, Y r.v.'s, a, b constants): $\mathrm{Var}(b) = 0$; $\mathrm{Var}(aX) = a^2\,\mathrm{Var}(X)$; $\mathrm{Var}(aX+b) = a^2\,\mathrm{Var}(X)$; if X, Y are independent, $\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$.

Example (archery): player A: $x = 8, 9, 10$ with $p(x) = 0.15, 0.40, 0.45$; player B: $y = 8, 9, 10$ with $p(y) = 0.35, 0.10, 0.55$. Then $\mathbb{E}(X) = 8(0.15) + 9(0.40) + 10(0.45) = 9.3$, $\mathbb{E}(Y) = 8(0.35) + 9(0.10) + 10(0.55) = 9.2$; $\mathrm{Var}(X) = (8-9.3)^2(0.15) + (9-9.3)^2(0.40) + (10-9.3)^2(0.45) = 0.51$, $\mathrm{Var}(Y) = (8-9.2)^2(0.35) + (9-9.2)^2(0.10) + (10-9.2)^2(0.55) = 0.86$. Alternatively $\mathrm{Var}(X) = 8^2(0.15) + 9^2(0.40) + 10^2(0.45) - 9.3^2 = 0.51$. (A has slightly higher mean and lower variance.) *Example (two flips):* $\mathbb{E}(X) = 1$, $\mathrm{Var}(X) = 0.5$.

HW #2, Q1 (Discrete Random Variables)

Problem: A discrete r.v. X has mean $\mathbb{E}(X) = \dfrac{9}{4}$ and cdf

$$F(x) = \begin{cases} 0, & x < 0, \\ c, & 0 \le x < 1, \\ c + \dfrac{1}{4}, & 1 \le x < 3, \\ \dfrac{7}{8}, & 3 \le x < 4, \\ 1, & x \ge 4. \end{cases}$$

(i) Find c and the pmf of X. (ii) Draw the graph of the cdf, marking open and closed endpoints clearly. (iii) Find $\mathbb{P}(1 \le X < 3)$ and $\mathbb{P}(1 \le X \le 3)$. (iv) Let $Y = 5 - 2X$. Find the mean and variance of Y.

Solution: (i) The pmf is recovered from the cdf by jumps, $p(x) = F(x) - F(x^-)$: $p(0) = c$, $p(1) = \left(c + \frac{1}{4}\right) - c = \frac{1}{4}$, $p(3) = \frac{7}{8} - \left(c + \frac{1}{4}\right)$, $p(4) = 1 - \frac{7}{8} = \frac{1}{8}$. Using the given mean, $\mathbb{E}(X) = 0 \cdot c + 1 \cdot \frac{1}{4} + 3\left(\frac{7}{8} - c - \frac{1}{4}\right) + 4 \cdot \frac{1}{8} = \frac{21}{8} - 3c = \frac{9}{4} \implies 3c = \frac{3}{8} \implies \boxed{c = \frac{1}{8}}$. Hence the pmf is $p(0) = \dfrac{1}{8}$, $p(1) = \dfrac{1}{4}$, $p(3) = \dfrac{1}{2}$, $p(4) = \dfrac{1}{8}$ (check: they sum to 1).

(ii) The cdf is a step function jumping at $x = 0, 1, 3, 4$ with jump heights $\dfrac{1}{8}, \dfrac{1}{4}, \dfrac{1}{2}, \dfrac{1}{8}$ respectively. Since F is right-continuous, each jump point is *closed* (filled dot) on the upper step and *open* (hollow dot) on the lower step: e.g., at $x = 1$ the point $\left(1, \dfrac{3}{8}\right)$ is filled while $\left(1, \dfrac{1}{8}\right)$ is hollow; between jumps the graph is flat, and $F = 1$ for all $x \ge 4$.

(iii) $\mathbb{P}(1 \le X < 3) = p(1) + p(2) = \frac{1}{4} + 0 = \frac{1}{4}$, $\mathbb{P}(1 \le X \le 3) = p(1) + p(2) + p(3) = \frac{3}{4}$. Equivalently via the cdf: $\mathbb{P}(1 \le X < 3) = F(3^-) - F(1^-) = \dfrac{3}{8} - \dfrac{1}{8} = \dfrac{1}{4}$ and $\mathbb{P}(1 \le X \le 3) = F(3) - F(1^-) = \dfrac{7}{8} - \dfrac{1}{8} = \dfrac{3}{4}$.

(iv) $\mathbb{E}(Y) = \mathbb{E}(5 - 2X) = 5 - 2 \cdot \dfrac{9}{4} = \dfrac{1}{2}$, and $\mathbb{E}(X^2) = 0^2 \cdot \frac{1}{8} + 1^2 \cdot \frac{1}{4} + 3^2 \cdot \frac{1}{2} + 4^2 \cdot \frac{1}{8} = \frac{27}{4} \implies$

$\mathrm{Var}(X) = \frac{27}{4} - \left(\frac{9}{4}\right)^2 = \frac{27}{16}$, so $\mathrm{Var}(Y) = \mathrm{Var}(5 - 2X) = (-2)^2 \mathrm{Var}(X) = 4 \cdot \dfrac{27}{16} = \dfrac{27}{4}$.

3.3 Continuous Random Variables

3.3.1 Probability Density Function (pdf)

For a continuous r.v. the possible values are numbers taken continuously on an interval: e.g., $\chi = [0, \infty)$, $\chi = [1, 3]$, $\chi = (2, 6]$. A continuous r.v. can be specified by its **cdf** $F(x) = \mathbb{P}(X \leq x)$ for any $x \in \mathbb{R}$, or by its **probability density function (pdf)** $f(x)$ satisfying: (i) $f(x) > 0$ for all $x \in \chi$ and $f(x) = 0$ for all $x \notin \chi$; (ii) $\displaystyle\int_{-\infty}^{\infty} f(x)\,dx = 1$.

Probability of an event: for $A \subseteq \mathbb{R}$, $\boxed{\mathbb{P}(X \in A) = \displaystyle\int_A f(x)\,dx}$ (the area under the density over A). For $A = (2, 4)$: $\displaystyle\int_2^4 f(x)\,dx$; for $B = \{2, 4\}$: the total area of the two dashed "lines" $= 0$. **In general, if X is continuous, $\mathbb{P}(X = k) = 0$ for any k**, so for any $a < b$: $\mathbb{P}(a < X \leq b) = \mathbb{P}(a \leq X \leq b) = \mathbb{P}(a \leq X < b) = \mathbb{P}(a < X < b)$. **Note:** the pdf value $f(x)$ does NOT give the probability that $X = x$; $f(x)$ measures how likely it is that X is *near* x.

3.3.2 Pdf and Cdf

For a continuous r.v. X: **pdf $\to$ cdf:** $F(a) = \mathbb{P}(X \leq a) = \displaystyle\int_{-\infty}^{a} f(x)\,dx$ for any real a (area under $y = f(x)$ up to a; $F(a)$ is the height of the cdf curve at a). $F(x)$ is non-decreasing *and continuous*. **cdf $\to$ pdf:** given $F(x)$, $f(a) = \dfrac{d}{dx}F(x)\Big|_{x=a}$ where the derivative exists. For any $a < b$: $\mathbb{P}(a < X \leq b) = F(b) - F(a) = \displaystyle\int_a^b f(x)\,dx$.

Example: $f(x) = \begin{cases} \dfrac{k}{x^2}, & x > 100 \\ 0, & x \leq 100 \end{cases}$. (i) By the pdf condition, $1 = \displaystyle\int_{100}^{\infty} \frac{k}{x^2}\,dx = \frac{k}{100} \implies k = 100$; and $F(x) = \begin{cases} 1 - \dfrac{100}{x}, & x > 100 \\ 0, & x \leq 100 \end{cases}$. (ii) $\mathbb{P}(50 < X \leq 1000) = \displaystyle\int_{100}^{1000} \frac{100}{x^2}\,dx = 100\left(\frac{1}{100} - \frac{1}{1000}\right) = \frac{9}{10}$.

3.3.3 Population Mean and Variance

Definition: if X has pdf $f(x)$, the population mean is $\boxed{\mu = \mathbb{E}(X) = \displaystyle\int_{-\infty}^{\infty} x f(x)\,dx}$, if convergent.

In general, $\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x)f(x)\,dx$.

Definition: the population variance is $\boxed{\mathrm{Var}(X) = \int_{-\infty}^{\infty} (x-\mu)^2 f(x)\,dx}$, if it exists; with $\mathrm{Var}(X) = \mathbb{E}[(X-\mu)^2] = \mathbb{E}(X^2) - \mu^2$; σ^2 (or σ_X^2) denotes it and $\sigma = \sqrt{\sigma^2}$ is the population sd.

Properties of $\mathbb{E}$ and Var (same as discrete): $\mathbb{E}(b) = b$, $\mathbb{E}(aX+b) = a\mathbb{E}(X) + b$, $\mathbb{E}(X \pm Y) = \mathbb{E}(X) \pm \mathbb{E}(Y)$; $\mathrm{Var}(b) = 0$, $\mathrm{Var}(aX+b) = a^2\,\mathrm{Var}(X)$; if X, Y independent, $\mathrm{Var}(X \pm Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$.

Example: $f(x) = \begin{cases} \frac{x^2}{3}, & -1 < x < 2 \\ 0, & \text{elsewhere} \end{cases}$; find $\mathbb{E}[g(X)]$ for $g(X) = 4X+3$: $\mathbb{E}(4X+3) = \int_{-1}^{2} (4x+3)\frac{x^2}{3}\,dx = \frac{1}{3}\int_{-1}^{2} (4x^3+3x^2)\,dx = \frac{1}{3}\left[x^4+x^3\right]_{-1}^{2} = 8$. *Remark:* also $\mathbb{E}(4X+3) = 4\,\mathbb{E}(X) + 3$.

Example (unit conversion): the mean temperature of a city is 50°F with sd 8°F; with $C = \frac{5}{9}(F-32)$: $\mathbb{E}(C) = \mathbb{E}\left(\frac{5}{9}F - \frac{160}{9}\right) = \frac{5}{9}\mathbb{E}(F) - \frac{160}{9} = \frac{5}{9}(50) - \frac{160}{9} = 10$°C, and $\sigma_C = \frac{5}{9}\sigma_F = \frac{5}{9}(8) = \frac{40}{9}$.

3.3.4 Chebyshev's Theorem

Theorem (Chebyshev's inequality): let X be a r.v. with mean μ and variance σ^2; then for any $t > 0$: $\boxed{\mathbb{P}(|X-\mu| \geq t) \leq \frac{\sigma^2}{t^2}}$.

Proof (continuous case; also holds for discrete): let $f(x)$ be the density of X; then $\mathbb{P}(|X-\mu| \geq t) = \int_{|x-\mu|\geq t} f(x)\,dx \leq \int_{|x-\mu|\geq t} \frac{(x-\mu)^2}{t^2} f(x)\,dx \leq \frac{1}{t^2}\int_{-\infty}^{\infty} (x-\mu)^2 f(x)\,dx = \frac{1}{t^2}\,\mathrm{Var}(X) = \frac{\sigma^2}{t^2}$.

Remark ($k\sigma$ form): letting $t = k\sigma$: $\mathbb{P}(|X-\mu| \geq k\sigma) \leq \frac{1}{k^2}$, or equivalently $\mathbb{P}(\mu - k\sigma < X < \mu + k\sigma) \geq 1 - \frac{1}{k^2}$: for $k=2$: $\geq 1 - \frac{1}{4} = \frac{3}{4}$; for $k=3$: $\geq 1 - \frac{1}{9} = \frac{8}{9}$.

Example: X has mean $\mu = 8$, variance $\sigma^2 = 9$, unknown distribution. (i) $\mathbb{P}(-4 < X < 20) = \mathbb{P}(8 - 4\sqrt{3} < X < 8 + 4\sqrt{3}) \geq 1 - \frac{1}{(4\sqrt{3}/3)^2} = \frac{15}{16}$. (ii) $\mathbb{P}(|X-8| \geq 6) = 1 - \mathbb{P}(-6 < X - 8 < 6) = 1 - \mathbb{P}(8 - 2\sqrt{3} < X < 8 + 2\sqrt{3}) \leq 1 - \left(1 - \frac{1}{(2\sqrt{3}/3)^2}\right) = \frac{1}{4}$.

Example (HW #2, Q5(ii)): X has mean 50, variance 16, unknown distribution. Find a lower bound for $\mathbb{P}(44 < X < 56)$. Rewrite the event as $\{|X-50| < 6\}$, so $t = 6$, $\sigma^2 = 16$: $\mathbb{P}(44 < X < 56) = 1 - \mathbb{P}(|X-50| \geq 6) \geq 1 - \frac{16}{6^2} = 1 - \frac{4}{9} = \frac{5}{9} \approx 0.5556$.

For comparison, when the distribution is known to be $N(50, 4^2)$, the exact probability is 0.8664 (see HW #2, Q5(i)) — Chebyshev is intentionally conservative because it must hold for *every* distribution with the given mean and variance.

Remarks: Chebyshev's theorem is distribution-free (holds for any distribution); only when the probability distribution is known can we determine exact probabilities.

3.3.5 Summary: Discrete vs. Continuous r.v.

	Discrete r.v.	**Continuous r.v.**
Possible values χ	isolated numbers, e.g., $\chi = \{0, 1, 2\}$, $\{1, 3, \dots\}$	numbers on an interval, e.g., $\chi = [0, \infty)$, $[1, 3]$
Distribution	pmf $p(x)$: probability; $0 < p(x) \le 1$; $\sum p(x) = 1$	pdf $f(x)$: density; $f(x) > 0$; $\int f = 1$
$\mathbb{P}(X \in A)$	$\mathbb{P}(X \in A) = \sum_{x \in A} p(x)$	$\mathbb{P}(X \in A) = \int_A f(x)\,dx$; $\mathbb{P}(X = k) = 0$
cdf	$F(a) = \mathbb{P}(X \le a)$; step function, jumps of size $p(a)$; increases only at possible values	$F(a) = \int_{-\infty}^{a} f(x)\,dx$; continuous; $f(a) = F'(a)$ if the derivative exists
Mean	$\mu = \mathbb{E}(X) = \sum_{\text{all } x} x\,p(x)$	$\mu = \mathbb{E}(X) = \int_{-\infty}^{\infty} x f(x)\,dx$
Variance	$\sigma^2 = \sum_{\text{all } x} (x-\mu)^2 p(x)$	$\sigma^2 = \int_{-\infty}^{\infty} (x-\mu)^2 f(x)\,dx$

Shared properties: $\mathbb{E}(g(X)) = \sum g(x)p(x)$ or $\int g(x)f(x)\,dx$; $\mathbb{E}(aX+b) = a\mathbb{E}(X) + b$; $\mathrm{Var}(aX+b) = a^2\,\mathrm{Var}(X)$; $\mathrm{Var}(X) = \mathbb{E}(X^2) - [\mathbb{E}(X)]^2$; $\mathbb{E}(X \pm Y) = \mathbb{E}(X) \pm \mathbb{E}(Y)$; if X, Y independent: $\mathrm{Var}(X \pm Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$.

HW #2, Q2 (Continuous Random Variables)

Problem: The nonnegative waiting time X, in minutes, until the next call satisfies $\mathbb{P}(X > x) = e^{-x/4}$ for $x \ge 0$. (i) Find the cdf and pdf of X for all real x. (ii) Find $\mathbb{E}(X)$ and $\mathrm{Var}(X)$ by integration. (iii) Show that $\mathbb{P}(X > s+t \mid X > s) = \mathbb{P}(X > t)$ for $s, t \ge 0$.

Solution: (i) X is nonnegative, so $F(x) = 0$ for $x < 0$; for $x \ge 0$, $F(x) = \mathbb{P}(X \le x) = 1 - \mathbb{P}(X > x) = 1 - e^{-x/4}$. Differentiating gives the pdf:

$$f(x) = \frac{d}{dx}F(x) = \begin{cases} \frac{1}{4}e^{-x/4}, & x \ge 0, \\ 0, & x < 0, \end{cases}$$

an exponential distribution with rate $\frac{1}{4}$.

(ii) By integration by parts, $\mathbb{E}(X) = \int_0^\infty x \cdot \frac{1}{4}e^{-x/4}\,dx = \left[-xe^{-x/4}\right]_0^\infty + \int_0^\infty e^{-x/4}\,dx = 4$, and integrating by parts twice, $\mathbb{E}(X^2) = \int_0^\infty x^2 \cdot \frac{1}{4}e^{-x/4}\,dx = 2 \cdot 4^2 = 32 \implies \mathrm{Var}(X) = 32 - 4^2 = 16$.

(iii) Since $\{X > s+t\} \subseteq \{X > s\}$, for $s, t \ge 0$: $\mathbb{P}(X > s+t \mid X > s) =$

$\dfrac{\mathbb{P}(X > s+t)}{\mathbb{P}(X > s)} = \dfrac{e^{-(s+t)/4}}{e^{-s/4}} = e^{-t/4} = \mathbb{P}(X > t)$. This is the *memoryless property*: having already waited s minutes does not change the distribution of the remaining waiting time.

3.4 Some Probability Distributions

3.4.1 0. Uniform Distribution

A uniform distribution models a r.v. that is equally likely to take any value in a fixed interval $[a, b]$: no subinterval is more or less likely than any other of the same length. It is the simplest continuous distribution.

pdf: $\boxed{f(x) = \begin{cases} \dfrac{1}{b-a}, & a \le x \le b, \\ 0, & \text{otherwise.} \end{cases}}$ **cdf:** $\boxed{F(x) = \begin{cases} 0, & x < a, \\ \dfrac{x-a}{b-a}, & a \le x \le b, \\ 1, & x \ge b. \end{cases}}$

Validity check: $f(x) > 0$ for $a \le x \le b$, and $\displaystyle\int_a^b \frac{1}{b-a}\,dx = 1$.

Mean and variance: the mean is the *midpoint*: $\mathbb{E}(X) = \displaystyle\int_a^b x \cdot \frac{1}{b-a}\,dx = \frac{1}{b-a} \cdot \frac{b^2-a^2}{2} = \frac{a+b}{2}$. For the variance, $\mathbb{E}(X^2) = \displaystyle\int_a^b x^2 \cdot \frac{1}{b-a}\,dx = \frac{1}{b-a} \cdot \frac{b^3-a^3}{3} = \frac{a^2+ab+b^2}{3}$, so $\mathrm{Var}(X) = \mathbb{E}(X^2) - [\mathbb{E}(X)]^2 = \dfrac{a^2+ab+b^2}{3} - \dfrac{(a+b)^2}{4} = \boxed{\dfrac{(b-a)^2}{12}}$.

The variance depends only on the *length* of the interval, not its location.

R corner: `dunif(x, min, max)`, `punif(q, min, max)`, `qunif(p, min, max)`, `runif(n, min, max)`.

```
dunif(3, min = 1, max = 7)   # pdf at x = 3: 1/6
punif(3, 1, 7)               # F(3) = (3-1)/(7-1) = 1/3
runif(5, min = 0, max = 1)   # 5 U(0,1) realizations
```

HW #2, Q4 (Uniform Distribution and Probability)

Problem: The service time X, in minutes, follows a uniform distribution over $[1, 7]$, with pdf $f(x) = \dfrac{1}{6}$ for $1 \le x \le 7$ and 0 otherwise. (i) Find $\mathbb{E}(X)$ and $\mathrm{Var}(X)$. (ii) Given $\mathbb{E}(Y) = \dfrac{4}{3}$, find c and $\mathrm{Var}(Y)$, where $1 < c < 7$ and

$$Y = \begin{cases} 0, & X \le c, \\ X - c, & X > c. \end{cases}$$

Solution: (i) By the formulas above, $\mathbb{E}(X) = \dfrac{1+7}{2} = 4$, $\mathrm{Var}(X) = \dfrac{(7-1)^2}{12} = 3$. Equivalently by integration: $\mathbb{E}(X) = \displaystyle\int_1^7 x \cdot \frac{1}{6}\,dx = 4$, $\mathbb{E}(X^2) = \displaystyle\int_1^7 x^2 \cdot \frac{1}{6}\,dx = 19 \implies \mathrm{Var}(X) = 19 - 16 = 3$.

(ii) For $1 < c < 7$, $\mathbb{E}(Y) = \int_c^7 (x-c) \cdot \frac{1}{6}\, dx = \frac{(7-c)^2}{12} = \frac{4}{3} \implies (7-c)^2 = 16 \implies$ $\boxed{c = 3}$, where $7 - c = 4 > 0$ because $1 < c < 7$. Then $\mathbb{E}(Y^2) = \int_3^7 (x-3)^2 \cdot \frac{1}{6}\, dx = \frac{4^3}{18} = \frac{32}{9} \implies \mathrm{Var}(Y) = \frac{32}{9} - \left(\frac{4}{3}\right)^2 = \frac{16}{9}$.

3.4.2 1. Bernoulli Random Variable

A Bernoulli r.v. takes only two values 0 and 1 with probabilities $1-p$ and p: $p(1) = p$, $p(0) = 1-p$, $p(x) = 0$ otherwise; equivalently $\boxed{p(x) = p^x(1-p)^{1-x}}$ for $x = 0, 1$ (and 0 otherwise). Any trial with two possible outcomes can be described by a Bernoulli r.v.: tossing a coin (head/tail); the grade of a P/F course (Pass/Fail); the gender of a newborn (boy/girl); a single shot (hit/miss). Its cdf: $F(x) = 0$ for $x < 0$, $F(x) = 1 - p$ for $0 \le x < 1$, $F(x) = 1$ for $x \ge 1$; and $\mathbb{E}(X) = p$, $\mathrm{Var}(X) = p(1-p)$.

3.4.3 2. Binomial Distribution

A Binomial distribution characterizes the number of successes in a random experiment with features: (1) a fixed finite number of identical trials, $n < \infty$; (2) trials are independent; (3) each trial results in two outcomes, *success* and *failure*; (4) the probability of success p is constant across trials. *Typical trials:* coin toss (head = success), pure guess in M.C. (correct = success), randomly selected product (non-defective = success).

Notation: $X \sim \mathrm{Binomial}(n, p)$ where $n =$ number of trials, $p =$ probability of success.

pmf: $\boxed{p(x) = \mathbb{P}(X = x) = \binom{n}{x} p^x (1-p)^{n-x}}$ for $x = 0, 1, \ldots, n$, and $p(x) = 0$ otherwise.

Derivation: x successes occur in n trials in $\binom{n}{x}$ ways; each such sequence has probability $p^x(1-p)^{n-x}$.

Remarks: if $n = 1$, X is a Bernoulli r.v.; validity check: $p(x) > 0$ for $x = 0, \ldots, n$ and $\sum_{x=0}^{n} \mathbb{P}(X = x) = \sum_{x=0}^{n} \binom{n}{x} p^x (1-p)^{n-x} = (p + 1 - p)^n = 1$ by the **Binomial Theorem** $(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k}$; **mean and variance:** $\mathbb{E}(X) = np$, $\mathrm{Var}(X) = np(1-p)$. The pmf shape depends on n and p (e.g., $n = 10$: $p = 0.5$ symmetric; $p = 0.1$ right-skewed).

Example (number "7"): three students each randomly pick a number between 0 and 9; $X =$ number of students who pick "7". Check features: $n = 3$ identical trials; independent choices; success = picking "7", failure otherwise; $p = \frac{1}{10}$ constant. Thus $X \sim \mathrm{Binomial}(3, 0.1)$, and with $Y =$ "7", $N =$ not "7": $p(0) = (0.1)^0(0.9)^3 = 0.729$; $p(1) = \binom{3}{1}(0.1)(0.9)^2 = 0.243$; $p(2) = \binom{3}{2}(0.1)^2(0.9) = 0.027$; $p(3) = (0.1)^3 = 0.001$.

R corner (binomial pmf and plot): for every distribution in R there are four functions,

distinguished by their first letter: `d` = density/pmf, `p` = cdf $F(x) = \mathbb{P}(X \le x)$, `q` = quantile (inverse cdf), `r` = random realizations. For the binomial they are `dbinom`, `pbinom`, `qbinom`, `rbinom` with arguments `size` and `prob`.

```
dbinom(0:3, size = 3, prob = 0.1)   # 0.729 0.243 0.027 0.001
pbinom(1, size = 3, prob = 0.1)     # P(X <= 1) = 0.972
pbinom(1, 3, 0.1, lower.tail = FALSE)  # P(X > 1) = 0.028
plot(0:3, dbinom(0:3, size = 3, prob = 0.1), type = "h",
     main = "Binomial Distribution (n=3, p=0.1)",
     ylab = "Probability", xlab = "# Successes", lwd = 3)
```

R corner (simulation): `rbinom()` generates realizations; set the random seed first for reproducible results. Comparing the relative frequency with the true probability ($\mathbb{P}(X > 1) = 0.028$) illustrates the law of large numbers.

```
set.seed(30)
z <- rbinom(500, size = 3, prob = 0.1)  # 500 realizations of X
table(z)          # count how many 0s, 1s, 2s, 3s occurred
mean(z > 1)       # relative frequency of {X > 1}, near 0.028
```

Example (insurance): 10,000 people buy life insurance; each dies within the following year with probability 0.005. Let X = total deaths: $X \sim \text{Binomial}(10000, 0.005)$. (i) $\mathbb{P}(X = 40) = \binom{10000}{40}(0.005)^{40}(0.995)^{9960} \approx 0.0214$; (ii) $\mathbb{P}(X \le 70) = \sum_{k=0}^{70} \binom{10000}{k}(0.005)^k(0.995)^{10000-k} \approx 0.997$.

```
choose(10000,40)*(0.005)^40*(0.995)^9960  # direct: 0.02143481
dbinom(40, size = 10000, prob = 0.005)    # same:   0.02143481
pbinom(70, size = 10000, prob = 0.005)    # P(X<=70): 0.997097
```

Example (with/without replacement): 5 defective products among 100; probability of getting 2 defectives in 3 draws: (i) *with replacement*: $X \sim \text{Binomial}(3, 0.05)$, $\mathbb{P}(X = 2) = \binom{3}{2}(0.05)^2(0.95) = 0.007125$; (ii) *without replacement*: Y does NOT follow a Binomial distribution (it is **Hypergeometric**): $\mathbb{P}(Y = 2) = \frac{\binom{5}{2}\binom{95}{1}}{\binom{100}{3}} \approx 0.005875$.

3.4.4 3. Poisson Distribution

The Poisson distribution determines probabilities of *counts* of the occurrence of an event over time (or space). *Typical examples:* the number of traffic accidents on a highway in a day; crashes of a computer network per week; the number of people joining a line in an hour; customers arrived per day; goals scored in a hockey game; typos per page of an essay; trees in an acre. **Features:** events are independent of each other (one occurrence does not affect the probability of another); the average rate (events per unit time period) is constant.

Notation: $X \sim \text{Poisson}(\lambda)$ where $\lambda \in (0, \infty)$ is the rate of occurrences per unit time (or space), i.e., the average number of occurrences per unit. **pmf:** $\boxed{p(x) = \mathbb{P}(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}}$ for $x = 0, 1, \ldots,$ and $p(x) = 0$ otherwise. *Validity*

check: $p(x) > 0$ and $\displaystyle\sum_{x=0}^{\infty} \mathbb{P}(X = x) = \sum_{x=0}^{\infty} \frac{\lambda^x e^{-\lambda}}{x!} = e^{-\lambda} \sum_{x=0}^{\infty} \frac{\lambda^x}{x!} = e^{-\lambda} \cdot e^{\lambda} = 1$, using $e^{\lambda} = 1 + \lambda + \dfrac{\lambda^2}{2!} + \dfrac{\lambda^3}{3!} + \cdots = \displaystyle\sum_{n=0}^{\infty} \frac{\lambda^n}{n!}$.

Mean and variance: $\mathbb{E}(X) = \displaystyle\sum_{k=0}^{\infty} k \cdot \frac{\lambda^k}{k!} e^{-\lambda} = \lambda e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!} = \lambda e^{-\lambda} \sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = \lambda$;

$\mathbb{E}(X^2) = \mathbb{E}[X(X-1)] + \mathbb{E}(X) = \displaystyle\sum_{k=0}^{\infty} k(k-1) \frac{\lambda^k}{k!} e^{-\lambda} + \lambda = \lambda^2 e^{-\lambda} \sum_{k=2}^{\infty} \frac{\lambda^{k-2}}{(k-2)!} + \lambda = \lambda^2 + \lambda$;

hence $\mathrm{Var}(X) = \mathbb{E}(X^2) - [\mathbb{E}(X)]^2 = \lambda^2 + \lambda - \lambda^2 = \lambda$. So $\boxed{\mathbb{E}(X) = \mathrm{Var}(X) = \lambda}$.

Example (flying bombs): from June 1944 to March 1945 Germany launched 9,251 V-1 "buzz" bombs against England; 537 struck South London. R.D. Clarke divided a 12 km × 12 km region into 576 squares and showed the pattern of hits follows a Poisson distribution. Since 537 bombs struck 576 regions, $\lambda = \dfrac{537}{576} = 0.9323$; with X = hits in a selected region, $X \sim \text{Poisson}(0.9323)$ and $\mathbb{P}(X = 2) = \dfrac{(0.9323)^2 e^{-0.9323}}{2!} \approx 0.1711$.

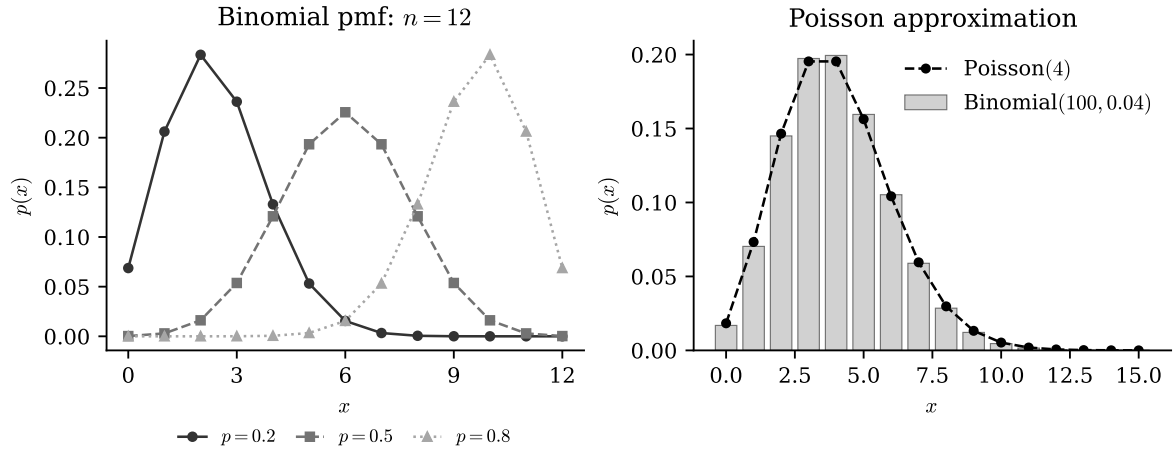
```
dpois(2, lambda = 0.9323)   # Poisson pmf at x = 2: 0.1710758
```

Example (estate agent): an agent sells 1.6 houses per day on average, counts follow a Poisson distribution. (a) selling 4 houses in a day: $X \sim \text{Poisson}(1.6)$, $\mathbb{P}(X = 4) = \dfrac{1.6^4 e^{-1.6}}{4!} = 0.055131$; (b) at least 5 in a day: $\mathbb{P}(X \geq 5) = 1 - \displaystyle\sum_{x=0}^{4} \frac{1.6^x e^{-1.6}}{x!} = 0.023682$; (c) selling 4 houses in two days: $Y \sim \text{Poisson}(3.2)$, $\mathbb{P}(Y = 4) = \dfrac{3.2^4 e^{-3.2}}{4!} = 0.178093$.
Remark: counting over a period of t units with rate λ per unit: use $Y_t \sim \text{Poisson}(\lambda t)$.

```
dpois(4, lambda = 1.6)            # (a) P(X = 4):       0.055131
1 - ppois(4, lambda = 1.6)        # (b) P(X >= 5):      0.023682
dpois(4, lambda = 3.2)            # (c) P(Y = 4):       0.178093
rpois(5, lambda = 1.6)            # 5 simulated daily sales counts
```

Relationship between Binomial and Poisson: the binomial tends to the Poisson as $n \to \infty$, $p \to 0$ with np constant. **Poisson limit theorem (Poisson approximation to Binomial):** if $p_n \in (0, 1)$ and $np_n \to \lambda$ (finite), then $\displaystyle\lim_{n\to\infty} \binom{n}{k} p_n^k (1 - p_n)^{n-k} = \frac{\lambda^k e^{-\lambda}}{k!}$.
The Poisson distribution with $\lambda = np$ closely approximates the binomial when n is large and p is small. *Insurance revisited in R* ($n = 10000$, $p = 0.005$, $\lambda = np = 50$) — the binomial and Poisson values agree to three decimal places:

```
dbinom(40, size = 10000, prob = 0.005)  # 0.02143481
dpois(40, lambda = 50)                  # 0.02149963
pbinom(70, size = 10000, prob = 0.005)  # 0.997097
ppois(70, lambda = 50)                  # 0.9970293
```



HW #2, Q3 (Binomial and Poisson Distributions)

Problem: Batch A contains 800 items, each defective with probability 0.005; Batch B contains 20 items, each defective with probability 0.2. Items within each batch are independent. Let X_A and X_B count the defective items. (i) State the distributions of X_A and X_B; calculate $\mathbb{P}(X_A \leq 1)$ and $\mathbb{P}(X_B \leq 1)$ separately. (ii) Use Poisson distributions to approximate the two probabilities in (i). (iii) For which batch is the Poisson approximation more accurate? Explain using the binomial means and variances. [Use R with appropriate functions to check the probabilities in (i) and (ii).]

Solution: (i) Both count successes in independent Bernoulli trials with constant p, so $X_A \sim \text{Binomial}(n = 800,\ p = 0.005)$, $X_B \sim \text{Binomial}(n = 20,\ p = 0.2)$. Then $\mathbb{P}(X_A \leq 1) = (0.995)^{800} + \binom{800}{1}(0.005)(0.995)^{799} \approx 0.0181 + 0.0729 = 0.0910$,

$$\mathbb{P}(X_B \leq 1) = (0.8)^{20} + \binom{20}{1}(0.2)(0.8)^{19} \approx 0.0115 + 0.0576 = 0.0692.$$

(ii) With $\lambda = np$: batch A gives $\lambda_A = 800 \times 0.005 = 4$, batch B gives $\lambda_B = 20 \times 0.2 = 4$. Using the Poisson approximation $Y \sim \text{Poisson}(4)$ for both: $\mathbb{P}(X \leq 1) \approx \mathbb{P}(Y \leq 1) = e^{-4} + \dfrac{4e^{-4}}{1!} = 5e^{-4} \approx 0.0916$ for both batches.

(iii) The approximation is far more accurate for **batch A** ($|0.0916 - 0.0910| \approx 0.0006$) than for batch B ($|0.0916 - 0.0692| \approx 0.0224$). A Poisson r.v. has mean = variance. For batch A, mean $= np = 4$ and variance $= np(1-p) = 3.98$ are nearly equal (large n, small p); for batch B, mean $= 4$ but variance $= 20 \times 0.2 \times 0.8 = 3.2$ is far from the mean. So X_A is well described by Poisson(4), while X_B is not.

```
pbinom(1, size = 800, prob = 0.005)   # exact batch A: 0.09102844
pbinom(1, size = 20,  prob = 0.2)     # exact batch B: 0.06917529
ppois(1, lambda = 4)                  # Poisson approx: 0.09157819
```

When is the Poisson approximation reliable?

The Poisson limit theorem requires n large and p small with $np = \lambda$ moderate. A quick check: compare the binomial variance $np(1-p)$ with the Poisson variance $\lambda = np$ — the closer $1-p$ is to 1, the better the approximation. Batch A: $1-p = 0.995$ (excellent); batch B: $1 - p = 0.8$ (poor).

How to determine λ in practice:

- *From the mean:* the most common method — since $\mathbb{E}(X) = \lambda$, estimate λ by the sample mean $\hat{\lambda} = \overline{x}$ (method of moments). The flying bombs example: $\hat{\lambda} = 537/576 = 0.9323$ per square.
- *From a Binomial approximation:* if $X \sim \text{Binomial}(n, p)$ with n large, p small, set $\lambda = np$ (HW #2, Q3 uses $\lambda = 800 \times 0.005 = 4$ and $\lambda = 20 \times 0.2 = 4$).
- *From the mgf:* if $\mathbb{E}(a^X)$ is known for some $a > 0$, then $\mathbb{E}(a^X) = M_X(\ln a) = e^{\lambda(a-1)}$, so $\lambda = \dfrac{\ln \mathbb{E}(a^X)}{a-1}$ (HW #2, Q6 with $a = 2$: $\mathbb{E}(2^X) = e^3 \implies \lambda = 3$).
- *From counting over t units:* if events occur at rate λ per unit, the count over t units is Poisson(λt); solve for λ if the rate over a different period is known (estate agent example: 1.6 houses/day $\implies \lambda = 1.6$).

3.4.5 4. Normal Distribution

The most important continuous distribution in statistics: it accurately describes many real-world quantities (noise, voltage, current, electrons, …) that result from many small independent random terms. Also called the **Gaussian distribution** (after Johann Carl Friedrich Gauss, "the Prince of Mathematics").

Notation: $X \sim N(\mu, \sigma^2)$ where $\mu \in (-\infty, \infty)$ and $\sigma \in (0, \infty)$. **pdf:** $\boxed{f(x) = \dfrac{1}{\sqrt{2\pi}\,\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}}$ for all real x. **Validity check:** $f(x) > 0$; and $\displaystyle\int_{-\infty}^{\infty} f(x)\,dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \frac{d(x-\mu)}{\sigma} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-t^2/2}\,dt = \frac{1}{\sqrt{2\pi}} \cdot \sqrt{2\pi} = 1.$ **Mean and variance:** $\mathbb{E}(X) = \displaystyle\int_{-\infty}^{\infty} x f(x)\,dx = \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} [(x-\mu) + \mu]\, e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx = \frac{\mu}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-t^2/2}\,dt + \frac{\sigma}{\sqrt{2\pi}} \int_{-\infty}^{\infty} t e^{-t^2/2}\,dt = \mu$ (with $t = \dfrac{x-\mu}{\sigma}$; the second integral is 0); $\text{Var}(X) = \displaystyle\int_{-\infty}^{\infty} (x-\mu)^2 f(x)\,dx = \frac{\sigma^2}{\sqrt{2\pi}} \int_{-\infty}^{\infty} t^2 e^{-t^2/2}\,dt = \frac{\sigma^2}{\sqrt{2\pi}} \int_{-\infty}^{\infty} -t\,d\left(e^{-t^2/2}\right) = \sigma^2$ (integration by parts). So $\mathbb{E}(X) = \mu$, $\text{Var}(X) = \sigma^2$.

All normal distributions have the **bell-shaped** density curve regardless of μ and σ; μ locates the center and σ controls the spread (larger σ, flatter curve with peak $\dfrac{1}{\sqrt{2\pi}\sigma}$).

Standard Normal distribution: the normal distribution with mean 0 and variance 1, $N(0, 1)$; its r.v. is usually denoted Z, with pdf and cdf $\varphi(z) = \dfrac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}$, $\Phi(z) = \displaystyle\int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}}\,dt$. By symmetry $\boxed{\Phi(-z_0) = 1 - \Phi(z_0)}$.

Standard Normal Table: the table gives $\Phi(z) = \mathbb{P}(Z \le z) = \displaystyle\int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-t^2/2}\,dt$, e.g., $\mathbb{P}(Z \le 0.25) = \Phi(0.25) = 0.5987$; $\mathbb{P}(Z \le -0.25) = 1 - 0.5987 = 0.4013 = \Phi(-0.25)$; $\mathbb{P}(Z > 0.25) = 1 - 0.5987 = 0.4013$.

Standardization: a process transforming any normal r.v. to standard normal. **If**

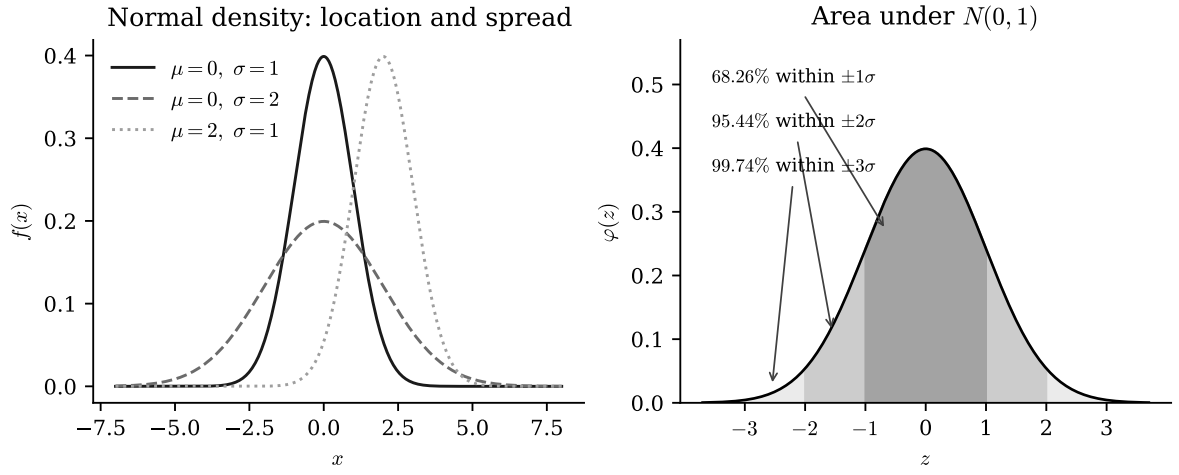
$X \sim N(\mu, \sigma^2)$, **then** $\boxed{Z = \dfrac{X-\mu}{\sigma} \sim N(0,1)}$: subtract μ, divide by σ. Consequently, the probability of any normal r.v. reduces to that of a standard normal r.v. (it suffices to know how to find probabilities of Z).

R corner (normal probabilities): `pnorm(q, mean, sd)` returns $\mathbb{P}(X \le q)$; note the argument is the standard deviation σ (not the variance σ^2), and `lower.tail = FALSE` gives an upper-tail probability. `dnorm` is the density, `qnorm` is the inverse cdf (percentiles), `rnorm` generates realizations.

```
pnorm(0.25)                          # Phi(0.25) = P(Z <= 0.25) = 0.5987
pnorm(-0.25)                         # Phi(-0.25) = 0.4013 by symmetry
1 - pnorm(0.25)                      # P(Z > 0.25)  = 0.4013
# X ~ N(1, 4), so sigma = 2; find P(X > 1.5):
pnorm(1.5, mean = 1, sd = 2, lower.tail = FALSE)   # 0.4013
qnorm(0.95)                          # z with Phi(z) = 0.95:  z = 1.6449
qnorm(0.975)                         # 1.96 (95% confidence intervals)
rnorm(5)                             # 5 realizations of Z ~ N(0, 1)
dnorm(0)                             # density at 0: 1/sqrt(2*pi) = 0.3989
```

Example: $X \sim N(1,4)$; find $\mathbb{P}(X > 1.5)$: $\mathbb{P}\left(\dfrac{X-1}{2} > \dfrac{1.5-1}{2}\right) = \mathbb{P}(Z > 0.25) = 1 - \mathbb{P}(Z \le 0.25) = 1 - 0.5987 = 0.4013$.

Example ($\pm k\sigma$ probabilities): $X \sim N(\mu, \sigma^2) \implies Z = \dfrac{X-\mu}{\sigma} \sim N(0,1)$, so $\mathbb{P}(\mu - \sigma < X \le \mu + \sigma) = \mathbb{P}(-1 < Z \le 1) = \Phi(1) - \Phi(-1) = 2\Phi(1) - 1 = 0.6826$; similarly $\mathbb{P}(\mu - 2\sigma < X \le \mu + 2\sigma) = 2\Phi(2) - 1 = 0.9544$ and $\mathbb{P}(\mu - 3\sigma < X \le \mu + 3\sigma) = 2\Phi(3) - 1 = 0.9974$.



Example (train routes): Route I (through town): travel time $X \sim N(50, 100)$ minutes; Route II (around town): $Y \sim N(60, 16)$ minutes. (a) With 70 minutes left: $\mathbb{P}(X \le 70) = \mathbb{P}\left(Z \le \dfrac{70-50}{10}\right) = \Phi(2) = 0.9773$ vs. $\mathbb{P}(Y \le 70) = \Phi\left(\dfrac{70-60}{4}\right) = \Phi(2.5) = 0.9938$; Route II is better. (b) With 65 minutes left (exercise): compare $\Phi(1.5) = 0.9332$ vs. $\Phi(1.25) = 0.8944$; Route I is better.

```
pnorm(70, mean = 50, sd = 10)   # Route I,   70 min: 0.97725
pnorm(70, mean = 60, sd = 4)    # Route II,  70 min: 0.99379
pnorm(65, mean = 50, sd = 10)   # Route I,   65 min: 0.93319
pnorm(65, mean = 60, sd = 4)    # Route II,  65 min: 0.89435
```

HW #2, Q5 (Normal Distribution and Chebyshev)

Problem: A measurement X follows a normal distribution with mean 50 and standard deviation 4. Let $\Phi(z)$ be the cdf of $Z \sim N(0,1)$, and use $\Phi(1.5) \approx 0.9332$. (i) Find $\mathbb{P}(44 < X < 56)$. (ii) Now suppose only the mean 50 and variance 16 are known, and the distribution is unspecified; use Chebyshev's inequality to give a lower bound for $\mathbb{P}(44 < X < 56)$. (iii) For $W \sim N(50, \sigma^2)$, find the largest $\sigma > 0$ such that $\mathbb{P}(W < 44) \leq \Phi(-1.645)$. [Use R to check your answer in (i).]

Solution: (i) Standardizing with $Z = \dfrac{X-50}{4}$: $\mathbb{P}(44 < X < 56) = \mathbb{P}\left(\dfrac{44-50}{4} < Z < \dfrac{56-50}{4}\right) = \mathbb{P}(-1.5 < Z < 1.5)$

$$= 2\Phi(1.5) - 1 = 2(0.9332) - 1 = 0.8664.$$

(ii) Chebyshev's inequality: for any r.v. with mean μ and variance σ^2, $\mathbb{P}(|X-\mu| \geq t) \leq \dfrac{\sigma^2}{t^2}$. Here $\{44 < X < 56\} = \{|X-50| < 6\}$, so $\mathbb{P}(44 < X < 56) = 1 - \mathbb{P}(|X - 50| \geq 6) \geq 1 - \dfrac{16}{6^2} = 1 - \dfrac{4}{9} = \dfrac{5}{9} \approx 0.5556$. The bound is much weaker than the normal answer 0.8664 — Chebyshev holds for *all* distributions with the given mean and variance, so it is conservative.

(iii) $\mathbb{P}(W < 44) = \Phi\left(\dfrac{44-50}{\sigma}\right) = \Phi\left(-\dfrac{6}{\sigma}\right)$. Since Φ is increasing, $\Phi\left(-\dfrac{6}{\sigma}\right) \leq \Phi(-1.645) \iff -\dfrac{6}{\sigma} \leq -1.645 \iff \dfrac{6}{\sigma} \geq 1.645 \iff \sigma \leq \dfrac{6}{1.645}$, so the largest value is $\sigma = \dfrac{6}{1.645} \approx 3.6474$.

```
pnorm(1.5) - pnorm(-1.5)              # (i) P(44 < X < 56): 0.86638
pnorm(56, 50, 4) - pnorm(44, 50, 4)   # same, without standardizing
```

3.5 Moment Generating Function (MGF)

3.5.1 Population Moments

Definition: the k-th (population) **moment about the origin** of (the distribution of) X is $\mathbb{E}(X^k)$, for $k = 1, 2, \ldots$, if it exists. The k-th (population) **central moment** is $\mathbb{E}\{[X - \mathbb{E}(X)]^k\}$, for $k = 1, 2, \ldots$, if it exists.

The first two moments are the most important: the population mean $\mu = \mathbb{E}(X)$ measures the *location* of a distribution; the population variance $\sigma^2 = \mathrm{Var}(X) = \mathbb{E}[(X-\mu)^2] = \mathbb{E}(X^2) - [\mathbb{E}(X)]^2$ measures the *spread* of a distribution.

Additionally, the 3rd and 4th population moments describe other characteristics of the distribution: **skewness** $\mathbb{E}\left[\left(\dfrac{X-\mu}{\sigma}\right)^3\right]$ (a variant of the 3rd population moment) and **kurtosis** $\mathbb{E}\left[\left(\dfrac{X-\mu}{\sigma}\right)^4\right]$ (a variant of the 4th population moment).

Skewness is the 3rd population moment of a standardized r.v.: $\mathbb{E}\left[\left(\dfrac{X-\mu}{\sigma}\right)^3\right] =$

$\frac{\mu_3}{\sigma^3}$, where μ_3 is the 3rd population central moment. It measures the *symmetry* of the distribution: a symmetric distribution (e.g., normal; more generally one symmetric about its median or mean) has skewness 0; a distribution with a long tail to the right has positive skewness; a long tail to the left gives negative skewness.

3.5.2 MGF of a Random Variable

Definition: the **moment generating function (mgf)** of X is $\boxed{M_X(t) = \mathbb{E}(e^{tX})}$, i.e., $M_X(t) = \sum_x e^{tx} p_X(x)$ if X is discrete with pmf $p_X(x)$, and $M_X(t) = \int_{-\infty}^{\infty} e^{tx} f_X(x)\,dx$ if X is continuous with pdf $f_X(x)$, provided the expectation exists and is finite for all real t around zero.

Remarks: (i) $M_X(t)$ is a function of t. (ii) Since the exponential function increases rapidly, the mgf does NOT always exist (e.g., the Cauchy distribution). (iii) When the mgf of a r.v. exists, we can use it to determine population moments in a very easy way. (iv) Throughout this course, unless otherwise specified, we assume that all mgfs exist and are finite for all real t around zero.

3.5.3 Use MGF to Find Population Moments

Assume that the interchange of the differentiation/summation and expectation operators is legitimate. Then, differentiating $M_X(t)$ repeatedly: $\frac{d}{dt}M_X(t) = \sum_x x e^{tx} p_X(x)$ (discrete) $= \int_{-\infty}^{\infty} x e^{tx} f_X(x)\,dx$ (continuous); $\frac{d^2}{dt^2}M_X(t) = \sum_x x^2 e^{tx} p_X(x)$ $= \int_{-\infty}^{\infty} x^2 e^{tx} f_X(x)\,dx$; and in general $\frac{d^k}{dt^k}M_X(t) = \sum_x x^k e^{tx} p_X(x) = \int_{-\infty}^{\infty} x^k e^{tx} f_X(x)\,dx$.

Evaluating the k-th derivative at $t = 0$ gives the relationship between the MGF and the k-th moment about the origin: $\boxed{M_X^{(k)}(0) = \frac{d^k}{dt^k}M_X(t)\Big|_{t=0} = \mathbb{E}(X^k)}$ for $k = 1, 2, \ldots$

Remark: $M_X^{(0)}(0) = M_X(0) = 1$.

3.5.4 MGF of Binomial Distribution

Consider $X \sim$ Binomial(n, p). We have $M_X(t) = \mathbb{E}(e^{tX}) = \sum_{k=0}^{n} e^{tk} p_X(k) = \sum_{k=0}^{n} e^{tk} \binom{n}{k} p^k (1-p)^{n-k} = \sum_{k=0}^{n} \binom{n}{k} (pe^t)^k (1-p)^{n-k} = (pe^t + 1 - p)^n$ by the Binomial Theorem $\left((a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k} \right)$.

Differentiating a first and second time yields $M_X'(t) = n(pe^t + 1 - p)^{n-1} pe^t$ and $M_X''(t) = n(n-1)(pe^t + 1 - p)^{n-2}(pe^t)^2 + n(pe^t + 1 - p)^{n-1} pe^t$.

So $\mathbb{E}(X) = M_X'(0) = np$ and $\mathbb{E}(X^2) = M_X''(0) = n(n-1)p^2 + np$. Thus, $\text{Var}(X) = \mathbb{E}(X^2) - [\mathbb{E}(X)]^2 = np(1-p)$.

Exercise: find the population mean and population variance for $X \sim \text{Poisson}(\lambda)$ using the mgf.

3.5.5 MGF of Normal Distribution

We first find the mgf of a standard normal distribution $Z \sim N(0,1)$: $M_Z(t) = \mathbb{E}(e^{tZ}) = \displaystyle\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{tz} e^{-\frac{z^2}{2}}\, dz = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(z^2 - 2tz)}\, dz$. Completing the square $z^2 - 2tz = (z-t)^2 - t^2$: $M_Z(t) = \dfrac{1}{\sqrt{2\pi}} \displaystyle\int_{-\infty}^{\infty} e^{-\frac{1}{2}[(z-t)^2 - t^2]}\, dz = e^{\frac{t^2}{2}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(z-t)^2}\, dz = e^{\frac{t^2}{2}} \displaystyle\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}u^2}\, du$ (substituting $u = z - t$), and the last integral is 1 (a normal pdf integrates to 1), so $\boxed{M_Z(t) = e^{\frac{t^2}{2}}}$ (also denoted $\exp\left(\dfrac{t^2}{2}\right)$).

Then $\mathbb{E}(Z) = M_Z'(0) = 0$, $\mathbb{E}(Z^2) = M_Z''(0) = 1$, and $\text{Var}(Z) = 1$.

Recall that for a r.v. $X \sim N(a, b)$ (with $a \in (-\infty, \infty)$, $b \in (0, \infty)$; pdf $f(x) = \dfrac{1}{\sqrt{2\pi b}} e^{-\frac{1}{2b}(x-a)^2}$ for all real x), we have $X = a + \sqrt{b}\, Z$. Thus $\mathbb{E}(X) = a + \sqrt{b}\, \mathbb{E}(Z) = a$, and $\text{Var}(X) = b\ \text{Var}(Z) = b$ — confirming the population mean a and population variance b of $N(a, b)$.

HW #2, Q6 (MGF of the Poisson Distribution)

Problem: Let $X \sim \text{Poisson}(\lambda)$, where $\lambda > 0$ is unknown, and suppose that $\mathbb{E}(2^X) = e^3$. (i) Derive the moment generating function $M_X(t)$ from its definition, in terms of λ. (ii) Use the mgf in (i) to determine λ. (iii) Find $M_X'(t)$ and $M_X''(t)$, and use these derivatives to calculate $\mathbb{E}(X)$ and $\text{Var}(X)$.

Solution: (i) By definition, valid for *all* $t \in (-\infty, \infty)$: $M_X(t) = \mathbb{E}(e^{tX}) = \displaystyle\sum_{x=0}^{\infty} e^{tx} \cdot \frac{e^{-\lambda}\lambda^x}{x!} = e^{-\lambda} \sum_{x=0}^{\infty} \frac{(\lambda e^t)^x}{x!} = e^{-\lambda} \cdot e^{\lambda e^t} = e^{\lambda(e^t - 1)}$, where the second-to-last step uses the Taylor series $e^u = \displaystyle\sum_{x=0}^{\infty} \frac{u^x}{x!}$ with $u = \lambda e^t$.

(ii) The key observation is $2^X = e^{X \ln 2}$, so $\mathbb{E}(2^X) = \mathbb{E}(e^{(\ln 2)X}) = M_X(\ln 2)$. Hence $e^3 = \mathbb{E}(2^X) = M_X(\ln 2) = e^{\lambda(e^{\ln 2} - 1)} = e^{\lambda} \implies \lambda = 3$.

(iii) Differentiating (chain rule, then product rule): $M_X'(t) = \lambda e^t\, e^{\lambda(e^t - 1)}$, $M_X''(t) = \left(\lambda e^t + \lambda^2 e^{2t}\right) e^{\lambda(e^t - 1)}$. Evaluating at $t = 0$ (so $e^{\lambda(e^0 - 1)} = e^0 = 1$): $\mathbb{E}(X) = M_X'(0) = \lambda = 3$, $\mathbb{E}(X^2) = M_X''(0) = \lambda + \lambda^2 = 3 + 9 = 12$, so $\text{Var}(X) = \mathbb{E}(X^2) - \mathbb{E}(X)^2 = 12 - 9 = 3$ — consistent with the general fact that a Poisson(λ) distribution has mean and variance both equal to λ.

Clever trick: turning $\mathbb{E}(a^X)$ into an mgf evaluation

Writing $a^X = e^{X \ln a}$ converts any raw moment $\mathbb{E}(a^X)$ into the mgf evaluated at $t = \ln a$ — no extra summation needed. This is exactly the technique used in the Exercise of Section 3.5.4 to find the population mean and variance of Poisson(λ) from its mgf.

4 Parameter Estimation

4.1 Basic Concepts

4.1.1 Motivation

In Chapter 3 we studied how to compute probabilities when the distribution of a r.v. X is fully known. In practice, however, the distribution is unknown, and so are its parameters (e.g., the population mean $\mu_X = \mathbb{E}(X)$ and population variance $\sigma_X^2 = \mathrm{Var}(X)$). **Inferential statistics** uses data from an observed sample to make conclusions about an unknown population.

We are interested in an **unknown population**: an unknown distribution of the r.v. X. We work with:

- A **parameter**: an unknown constant that indexes the distribution of X, e.g., μ_X and σ_X^2.
- A **sample**: a collection of observed values of X, denoted $x_1, x_2, \ldots, x_n$, where n is the **sample size**.
- A **statistic**: any function of the sample $T(x_1, x_2, \ldots, x_n)$ that does not depend on unknown parameters, e.g., $\overline{x}$, s_{n-1}^2, s_n^2.

4.1.2 Estimator vs. Estimate

The sample mean can be viewed in two roles:

- **As a number (estimate):** $\overline{x} = \dfrac{1}{n}\sum_{i=1}^{n} x_i$, where x_i is the i-th observed value. This is a realized number computed from a particular dataset.
- **As a random variable (estimator):** $\overline{X} = \dfrac{1}{n}\sum_{i=1}^{n} X_i$, where $X_1, X_2, \ldots, X_n$ are n copies of X, independent and identically distributed (i.i.d.). Before observation, $\overline{X}$ is a r.v. whose actual values are the sample means from different datasets; after observation, its value is $\overline{x}$.

Concretely, suppose we draw B different samples, each of size n, from the same population. The j-th sample gives n observed values $x_{1j}, \ldots, x_{nj}$ and one sample mean $\overline{x}_j$. The B numbers $\overline{x}_1, \overline{x}_2, \ldots, \overline{x}_B$ behave just like the original data $x_1, \ldots, x_n$, so we may regard them as the actual values of a new random variable, naturally denoted $\overline{X}$. This is the passage from a number to a r.v. Note that the unknown parameter μ_X is *fixed*, while the statistic $\overline{x}$ *changes* from sample to sample.

Example (light bulbs): A large batch of light bulbs has lifespan $X \sim N(\mu_X, \sigma_X^2)$ (the population). Five bulbs are sampled and their lifespans are $980, 960, 1030, 1300, 850$ days. Here $X_1, \ldots, X_5$ are the 5 copies with $X_i \sim N(\mu_X, \sigma_X^2)$ (i.i.d. = independent and identically distributed), the numbers $980, \ldots, 850$ are their observed values (data), and $\overline{x} = 5120/5 = 1024$ days is the estimate of μ_X.

	Estimator (methodology)	Estimate (numerical value)
Sample mean	$\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$	$\overline{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$
Sample variance	$S_{n-1}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2$	$s_{n-1}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \overline{x})^2$

Definition: any function of the random variables constituting a random sample is called a **statistic** if it does not depend on the unknown parameter(s). A statistic used to guess a parameter is an **estimator**; its observed value is an **estimate**.

4.1.3 Random Sample and Statistics

A **random sample** $X_1, X_2, \ldots, X_n$ from a population with cdf $F(x;\theta)$ (where θ denotes the unknown parameter(s)) satisfies:

- *Independence:* X_i does not affect any other X_j.
- *Identical distribution:* each X_i has the same distribution as the population, $X_i \sim F(x;\theta)$.

Common statistics (written as estimators, i.e., functions of the r.v.'s):

- Sample mean: $\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$.
- Sample variance (divide by $n-1$): $S_{n-1}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2$.
- Sample variance (divide by n): $S_n^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})^2$.
- Sample sd: $S_{n-1} = \sqrt{S_{n-1}^2}$ or $S_n = \sqrt{S_n^2}$.
- k-th sample moment about the origin: $\overline{X^k} = \frac{1}{n}\sum_{i=1}^{n} X_i^k$ $(k = 1, 2, \ldots)$.
- k-th sample central moment: $\frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})^k$ $(k = 1, 2, \ldots)$.

4.1.4 Sample Moment vs. Population Moment

The k-th **population moment about the origin** is $\mathbb{E}(X^k)$:

$$\mathbb{E}(X^k) = \begin{cases} \sum_x x^k p_X(x), & X \text{ discrete}, \\ \int_{-\infty}^{\infty} x^k f_X(x)\,dx, & X \text{ continuous}. \end{cases}$$

The k-th **sample moment about the origin** is $\overline{X^k} = \dfrac{1}{n}\sum_{i=1}^{n} X_i^k$.

Since $X_1^k, X_2^k, \ldots, X_n^k$ are i.i.d. with the same distribution as X^k, $\mathbb{E}(\overline{X^k}) = \dfrac{1}{n}\sum_{i=1}^{n}\mathbb{E}(X_i^k) = \dfrac{1}{n}\cdot n\,\mathbb{E}(X^k) = \mathbb{E}(X^k)$. Hence the sample moment is an unbiased estimator of the corresponding population moment: $\boxed{\mathbb{E}(\overline{X^k}) = \mathbb{E}(X^k)}$.

4.1.5 Numerical Description of $\overline{X}$ and S_{n-1}^2

Theorem: Assume the population has mean $\mathbb{E}(X) = \mu_X$ and variance $\mathrm{Var}(X) = \sigma_X^2$. Let $X_1, X_2, \ldots, X_n$ be a random sample. Then $\boxed{\mathbb{E}(\overline{X}) = \mu_X}$, $\boxed{\mathrm{Var}(\overline{X}) = \sigma_X^2/n}$, $\boxed{\mathbb{E}(S_{n-1}^2) = \sigma_X^2}$.

Proof: **Mean.** $\mathbb{E}(\overline{X}) = \mathbb{E}\left(\dfrac{1}{n}\sum_{i=1}^{n} X_i\right) = \dfrac{1}{n}\sum_{i=1}^{n}\mathbb{E}(X_i) = \dfrac{1}{n}\cdot n\mu_X = \mu_X$.

Variance. By independence, $\mathrm{Var}(\overline{X}) = \mathrm{Var}\left(\dfrac{1}{n}\sum_{i=1}^{n} X_i\right) = \dfrac{1}{n^2}\sum_{i=1}^{n}\mathrm{Var}(X_i) = \dfrac{1}{n^2}\cdot n\sigma_X^2 = \dfrac{\sigma_X^2}{n}$.

Sample variance. We start from the algebraic identity $(n-1)S_{n-1}^2 = \sum_{i=1}^{n}(X_i-\overline{X})^2 = \sum_{i=1}^{n}(X_i-\mu_X)^2 - n(\overline{X}-\mu_X)^2$, which follows from expanding $(X_i-\overline{X}) = (X_i-\mu_X)-(\overline{X}-\mu_X)$ and using $\sum(X_i-\overline{X}) = 0$. Taking expectation:

$$\begin{aligned}(n-1)\mathbb{E}(S_{n-1}^2) &= \sum_{i=1}^{n}\mathbb{E}[(X_i-\mu_X)^2] - n\mathbb{E}[(\overline{X}-\mu_X)^2] \\ &= \sum_{i=1}^{n}\sigma_X^2 - n\cdot\frac{\sigma_X^2}{n} = n\sigma_X^2 - \sigma_X^2 = (n-1)\sigma_X^2.\end{aligned}$$

Dividing both sides by $n-1$ gives $\mathbb{E}(S_{n-1}^2) = \sigma_X^2$. ■

```
# R corner: B = 5000 samples, size n = 10, Binomial(1, p = 0.7)
B <- 5000; n <- 10; p <- 0.7
```

```
xbars <- replicate(B, mean(rbinom(n, 1, p)))   # 5000 means
s2n1  <- replicate(B, var(rbinom(n, 1, p)))    # 5000 variances
mean(xbars); var(xbars); mean(s2n1)   # p, p(1-p)/n, p(1-p)
# Theory: mu = p = 0.7, sigma^2 = 0.21, Var(Xbar) = 0.021,
#         E(S_{n-1}^2) = 0.21
```

4.2 Point Estimation

4.2.1 General Concepts

Motivation. If the parameter of interest is a population mean (or population variance), the sample mean (or sample variance) may be used to estimate it. For example, from a batch of products n items are picked and k are found defective — what is the defective rate of the batch? Or 10 products have lifespans $1010, 980, 975, 1050, 1100, 990, 1020, 1150, 1210, 960$ hours — how do we estimate the average lifespan of the batch? For other parameters the question is harder: if claim amounts in insurance follow $f(x;\alpha,\theta) = \dfrac{1}{\theta}e^{-(x-\alpha)/\theta}$ for $x > \alpha$ and 0 otherwise, how do we estimate α and θ?

Let the pdf/pmf of X be $f(x;\vec{\theta})$ with unknown vector $\vec{\theta} = (\theta_1, \theta_2, \ldots, \theta_m)^\top \in \Theta \subseteq \mathbb{R}^m$, where Θ is the **parameter space** and $m \geq 1$ is the number of unknown parameters. Typical parameter spaces:

- $X \sim \text{Poisson}(\lambda)$: $\Theta = \{\lambda : \lambda > 0\}$.
- $X \sim N(\mu, \sigma^2)$: $\Theta = \{(\mu, \sigma^2)^\top : -\infty < \mu < \infty,\ \sigma > 0\}$.
- Exam scores $X \sim N(\mu, \sigma^2)$ with $0 \leq \mu \leq 100$: $\Theta = \{(\mu, \sigma^2)^\top : 0 \leq \mu \leq 100,\ 0 < \sigma < 100\}$.

We thus have a family of densities $\{f(x;\vec{\theta}) : \vec{\theta} \in \Theta\}$ and need to select one member of the family as the pdf of X, which is equivalent to estimating $\vec{\theta}$.

A **point estimator** is a statistic $\hat{\vec{\theta}} = T(X_1, X_2, \ldots, X_n)$ used to estimate $\vec{\theta}$; its observed value $\hat{\vec{\theta}} = T(x_1, x_2, \ldots, x_n)$ is a **point estimate**.

- Example: $\overline{X}$ estimates μ_X, and S_{n-1}^2 estimates σ_X^2.
- Why "point"? Because $T(x_1, \ldots, x_n)$ is a single point in $\mathbb{R}^m$ used to guess $\vec{\theta}$, another point in $\mathbb{R}^m$.
- Point estimators are not unique. For σ_X^2, S_{n-1}^2 is unbiased while S_n^2 is biased (but consistent).
- The parameter of interest (the **estimand**) may be a function $h(\vec{\theta})$ of the unknown parameters: for instance μ_X^2 instead of μ_X, or the coefficient of variation σ_X/μ_X instead of μ_X or σ_X alone.

4.2.2 Method of Moments Estimation (MME)

The **method of moments** (dating back to Karl Pearson, 1890s) estimates parameters by matching population moments with their sample counterparts.

Definition (MME): suppose there are m unknown parameters $\theta_1, \theta_2, \ldots, \theta_m$. If we

can write them as functions of the first m (or more) population moments,

$$\begin{aligned}\theta_1 &= g_1\big(\mathbb{E}(X), \mathbb{E}(X^2), \dots, \mathbb{E}(X^m), \dots\big),\\ \theta_2 &= g_2\big(\mathbb{E}(X), \mathbb{E}(X^2), \dots, \mathbb{E}(X^m), \dots\big),\\ &\vdots\\ \theta_m &= g_m\big(\mathbb{E}(X), \mathbb{E}(X^2), \dots, \mathbb{E}(X^m), \dots\big),\end{aligned}$$

then the **method of moments estimators** are obtained by replacing each population moment by the corresponding sample moment:

Method of moments estimators

$$\begin{aligned}\hat\theta_1 &= g_1\big(\overline{X}, \overline{X^2}, \dots, \overline{X^m}, \dots\big),\\ \hat\theta_2 &= g_2\big(\overline{X}, \overline{X^2}, \dots, \overline{X^m}, \dots\big),\\ &\vdots\\ \hat\theta_m &= g_m\big(\overline{X}, \overline{X^2}, \dots, \overline{X^m}, \dots\big).\end{aligned}$$

Equivalent procedure: set the first m sample moments equal to the first m population moments and solve the resulting system for $\hat\theta_1, \dots, \hat\theta_m$.

Three-step procedure: (1) find the first m population moments in terms of $\theta_1, \dots, \theta_m$; (2) invert these equations to express the parameters in terms of the moments; (3) substitute the sample moments.

MME Examples

Example 1: $X \sim N(\mu_X, \sigma_X^2)$, both unknown. We have $\mathbb{E}(X) = \mu_X$ and $\mathbb{E}(X^2) = \mathrm{Var}(X) + [\mathbb{E}(X)]^2 = \sigma_X^2 + \mu_X^2$.
Solving for the parameters: $\mu_X = \mathbb{E}(X)$ and $\sigma_X^2 = \mathbb{E}(X^2) - [\mathbb{E}(X)]^2$. Substituting sample moments, $\boxed{\hat\mu_{X,\mathrm{MME}} = \overline{X}}$, $\boxed{\hat\sigma^2_{X,\mathrm{MME}} = \overline{X^2} - \overline{X}^2 = S_n^2}$.
The same answer is reached by three equivalent routes:

- (I) invert the two moment equations and substitute the sample moments;
- (II) set $\mathbb{E}(X) = \overline{X}$ and $\mathbb{E}(X^2) = \overline{X^2}$, then solve the system for $(\hat\mu_X, \hat\sigma_X^2)$;
- (III) match the first moment about the origin $\mathbb{E}(X) = \overline{X}$ and the second central moment $\mathrm{Var}(X) = \dfrac{1}{n}\displaystyle\sum_{i=1}^{n}(X_i - \overline{X})^2$.

The algebra behind route (III) is $\dfrac{1}{n}\sum(X_i - \overline{X})^2 = \dfrac{1}{n}\sum X_i^2 - 2\overline{X}\cdot\dfrac{1}{n}\sum X_i + \overline{X}^2 = \overline{X^2} - \overline{X}^2$.
Example 1′: $X \sim N(\mu_X, 2411)$ with only μ_X unknown. Since $\mathbb{E}(X) = \mu_X$, we have $\mu_X = \mathbb{E}(X)$ and $\hat\mu_{X,\mathrm{MME}} = \overline{X}$. We only estimate the *unknown* parameter: for a known parameter such as the variance 2411 there is no point in estimating it.
Example 1″: $X \sim N(2411, \sigma_X^2)$ with only σ_X^2 unknown. Since $\mathbb{E}(X^2) = \mathrm{Var}(X) + [\mathbb{E}(X)]^2 = \sigma_X^2 + 2411^2$, we have $\sigma_X^2 = \mathbb{E}(X^2) - 2411^2$ and $\hat\sigma^2_{X,\mathrm{MME}} = \overline{X^2} - 2411^2 = \dfrac{1}{n}\displaystyle\sum_{i=1}^{n} X_i^2 - 2411^2$.
Example 2: $X \sim \mathrm{Poisson}(\lambda)$, λ unknown. We have $\mathbb{E}(X) = \lambda$, so $\hat\lambda_{\mathrm{MME}} = \overline{X}$.

Alternatively, since $\mathrm{Var}(X) = \lambda$ as well, we could also use $\hat{\lambda}_{\mathrm{MME}} = S_n^2$. The MME is *not always unique*; in this course we use the lowest-order moments to resolve this.
Example 3: Let $x_1 = 0.42, x_2 = 0.10, x_3 = 0.65, x_4 = 0.23$ be a sample from $f(x;\theta) = \theta x^{\theta-1}$ for $0 \le x \le 1$ (Beta-like). Find the MME of θ. $\mathbb{E}(X) = \int_0^1 x \cdot \theta x^{\theta-1}\,dx = \dfrac{\theta}{\theta+1}$. Setting equal to $\overline{x} = 0.35$: $\dfrac{\theta}{\theta+1} = 0.35 \implies \hat{\theta}_{\mathrm{MME}} = \dfrac{\overline{x}}{1-\overline{x}} = \dfrac{0.35}{0.65} \approx 0.538$.

Remarks on MME:

- Quick and easy, and defined by replacing population moments with sample moments; the MME may not be unique.
- Heavily relies on the existence of the required population moments, and works well only when the sample size is sufficiently large.
- Often yields biased estimators; does not use all the information in the data.
- If $\hat{\theta}_i$ is the MME of θ_i, then $h(\hat{\theta}_1, \ldots, \hat{\theta}_m)$ is the MME of $h(\theta_1, \ldots, \theta_m)$ for any known function h.

4.2.3 Goodness of Point Estimators

Point estimators are not unique; we need criteria to judge them. The classical criteria are unbiasedness, efficiency, consistency, sufficiency and completeness; here we focus on the first two.

Unbiasedness. An estimator $\hat{\theta}$ is **unbiased** for θ if $\mathbb{E}(\hat{\theta}) = \theta$ for all $\theta \in \Theta$. Otherwise it is **biased**, and $\mathrm{bias}_n(\hat{\theta}) = \mathbb{E}(\hat{\theta}) - \theta$ is the **bias**. If $\lim_{n\to\infty} \mathrm{bias}_n(\hat{\theta}) = 0$, the estimator is **asymptotically unbiased**.

From the theorem in 4.1, $\overline{X}$ is unbiased for μ_X and S_{n-1}^2 is unbiased for σ_X^2. By contrast, $S_n^2 = \dfrac{n-1}{n} S_{n-1}^2$ has $\boxed{\mathbb{E}(S_n^2) = \dfrac{n-1}{n}\sigma_X^2}$, so it is biased but asymptotically unbiased. This is why we use $n-1$ in the denominator for the sample variance.

Efficiency. Let $\hat{\theta}_1$ and $\hat{\theta}_2$ both be unbiased estimators of θ. If $\mathrm{Var}(\hat{\theta}_1) \le \mathrm{Var}(\hat{\theta}_2)$ for all $\theta \in \Theta$, then $\hat{\theta}_1$ is **more efficient** than $\hat{\theta}_2$. Efficiency measures *precision*: smaller variance means estimates are closer to each other across different samples. For example, $\overline{X}$ is more efficient than X_1 (for $n \ge 2$) since $\mathrm{Var}(\overline{X}) = \sigma_X^2/n < \sigma_X^2 = \mathrm{Var}(X_1)$.

Many unbiased estimators may exist, so a comparison is necessary. For any population with $\mathbb{E}(X) = \mu_X$, the statistics X_1, X_n and $\overline{X}$ are all unbiased for μ_X. For $X \sim \mathrm{Poisson}(\lambda)$, both $\overline{X}$ and S_{n-1}^2 are unbiased for λ, and so is every convex combination $\hat{\lambda} = c\,\overline{X} + (1-c)S_{n-1}^2$ with $c \in [0,1]$.

Unbiasedness alone therefore cannot choose an estimator, which is exactly why efficiency is introduced. Two further notions are useful:

- **Accuracy**: how close the estimator is to the unknown parameter.
- **Precision**: how close repeated estimates are to each other across samples.

Unbiasedness and Efficiency Example

Let $X_1, X_2, \ldots, X_n$ $(n \geq 2)$ be a random sample from

$$f(x;\theta) = \begin{cases} \dfrac{2x}{\theta^2}, & 0 \leq x \leq \theta, \\ 0, & \text{otherwise}, \end{cases}$$

where $\theta > 1$ is unknown.

(i) **MME.** Compute $\mathbb{E}(X) = \displaystyle\int_0^\theta x \cdot \frac{2x}{\theta^2}\, dx = \frac{2\theta}{3}$. So $\theta = \frac{3}{2}\mathbb{E}(X)$, and $\hat{\theta}_{\text{MME}} = \frac{3}{2}\overline{X}$.

(ii) **Unbiasedness.** $\mathbb{E}(\hat{\theta}_{\text{MME}}) = \frac{3}{2}\mathbb{E}(\overline{X}) = \frac{3}{2} \cdot \frac{2\theta}{3} = \theta$, so $\hat{\theta}_{\text{MME}}$ is unbiased.

(iii) **Variance.** $\mathbb{E}(X^2) = \displaystyle\int_0^\theta x^2 \cdot \frac{2x}{\theta^2}\, dx = \frac{\theta^2}{2}$, so $\text{Var}(X) = \frac{\theta^2}{2} - \left(\frac{2\theta}{3}\right)^2 = \frac{\theta^2}{18}$. Hence $\text{Var}(\hat{\theta}_{\text{MME}}) = \left(\frac{3}{2}\right)^2 \text{Var}(\overline{X}) = \frac{9}{4} \cdot \frac{1}{n} \cdot \frac{\theta^2}{18} = \frac{\theta^2}{8n}$.

(iv) Let $\hat{\theta}' = \frac{1}{2}X_1 + aX_2$. For unbiasedness, need $\mathbb{E}(\hat{\theta}') = \left(\frac{1}{2} + a\right)\frac{2\theta}{3} = \theta \implies a = 1$. Then $\hat{\theta}' = \frac{1}{2}X_1 + X_2$ is unbiased, and $\text{Var}(\hat{\theta}') = \frac{1}{4}\text{Var}(X_1) + \text{Var}(X_2) = \frac{5}{4} \cdot \frac{\theta^2}{18} = \frac{5\theta^2}{72} > \frac{\theta^2}{8n} = \text{Var}(\hat{\theta}_{\text{MME}})$ $(n \geq 2)$. So $\hat{\theta}_{\text{MME}} = \frac{3}{2}\overline{X}$ is more efficient than $\hat{\theta}' = \frac{1}{2}X_1 + X_2$.

4.3 Interval Estimation

4.3.1 Motivation

A point estimate provides a single guess for θ, but does not tell us about the *precision* or *reliability* of that guess. An **interval estimate** is a pair of statistics $T_1(X_1, \ldots, X_n) \leq T_2(X_1, \ldots, X_n)$ such that the random interval $[T_1, T_2]$ contains θ with a prescribed probability $1 - \alpha$.

Definition (confidence interval): for $\alpha \in (0, 1)$, if there exist statistics $T_1 \leq T_2$ such that $\mathbb{P}(T_1 \leq \theta \leq T_2) \geq 1 - \alpha$ for all $\theta \in \Theta$, then $[T_1, T_2]$ is a **random interval estimator** of θ. An observed value $[t_1, t_2]$ of $[T_1, T_2]$ is a $\mathbf{100(1-\alpha)\%}$ **confidence interval** for θ. The quantity $1 - \alpha$ is the **confidence level**.

The interval is *random* (endpoints are statistics, functions of the sample), while θ is *fixed*. The probability statement refers to the method: if we repeated the sampling procedure many times and computed a new interval each time, about $100(1 - \alpha)\%$ of these intervals would contain θ.

Point estimators share a fundamental weakness: they cannot provide the *precision* or the *reliability* of the estimate. For instance, let $X \sim \text{Poisson}(\lambda)$; a sample of size six gives the moment estimate $\hat{\lambda} = \overline{x} = 6.8$. A point estimate alone does not tell us whether the true λ is likely to lie in $[6.7, 6.9]$, or whether the estimation is so imprecise that λ could be as small as 1.0 or as large as 12.0.

Sometimes a range is more meaningful than a point: the sample means $\overline{x}$ are not the same for different samples, and in practice we quote a range (the height or age of a suspect, the cement consumption in constructing a building).

	Point estimation	Interval estimation
Methodology (r.v.)	estimator	random interval
Numerical (data)	estimate	confidence interval

There is a trade-off between precision and reliability: if $T_2 - T_1$ is small the precision is high but the reliability is low, and if $T_2 - T_1$ is large the reliability is high but the precision is low. The confidence level $1 - \alpha$ balances the two.

Remarks:

- α is usually small (0.01, 0.05 or 0.10) so that $1 - \alpha$ is close to 1 (0.99, 0.95 or 0.90); T_1 and T_2 are the **lower and upper confidence limits**.
- For a continuous population the interval is chosen so that $\mathbb{P}(T_1 \leq \theta \leq T_2) = 1 - \alpha$ exactly; for a discrete population we can only require $\mathbb{P}(T_1 \leq \theta \leq T_2) \geq 1 - \alpha$, as close to $1 - \alpha$ as possible.

A confidence interval is built from the **sampling distribution** of a statistic $T(X_1, \ldots, X_n)$, i.e. its probability distribution before observation. It depends heavily on the distribution of the original r.v. X. In this section we only treat a normal population $X \sim N(\mu_X, \sigma_X^2)$, for which two distributions (χ^2 and t) and three theorems (the sampling distributions of $\overline{X}$ and S_{n-1}^2) suffice.

4.3.2 Constructing a Random Interval

The recipe starts from a point estimator and its sampling distribution. Since $\mathbb{E}(\overline{X}) = \mu_X$, $\overline{X}$ is an unbiased point estimator of μ_X, so it is natural to center the interval at $\overline{X}$.

Given α, look for $k_1 > 0$ and $k_2 > 0$ such that $\mathbb{P}(\mu_X - k_1 \leq \overline{X} \leq \mu_X + k_2) = 1 - \alpha$. Subtracting μ_X and then $\overline{X}$ from all three parts (both operations are reversible), the event is the same as $\mathbb{P}(\overline{X} - k_2 \leq \mu_X \leq \overline{X} + k_1) = 1 - \alpha$.

The endpoints $\overline{X} - k_2$ and $\overline{X} + k_1$ are statistics, hence random, while the unknown constant μ_X is trapped between them: this is exactly the random interval we are looking for. If the distribution of $\overline{X}$ is known, k_1 and k_2 can be computed.

Symmetric choice. Taking $k_1 = k_2 = k$ gives $\mathbb{P}(|\overline{X} - \mu_X| \leq k) = 1 - \alpha$. If $\overline{X} \sim N(\mu_X, \sigma_X^2/n)$, then $k = z_{\alpha/2}\,\sigma_X/\sqrt{n}$. Numerically, for $\alpha = 0.05$, using $k = 2\sigma_X/\sqrt{n}$ gives $\mathbb{P}\left(\mu_X - 2\dfrac{\sigma_X}{\sqrt{n}} \leq \overline{X} \leq \mu_X + 2\dfrac{\sigma_X}{\sqrt{n}}\right) = 0.9544997$, which overshoots 0.95, whereas $\mathbb{P}\left(\mu_X - 1.96\dfrac{\sigma_X}{\sqrt{n}} \leq \overline{X} \leq \mu_X + 1.96\dfrac{\sigma_X}{\sqrt{n}}\right) = 0.9500042$, so 1.96 is used instead of 2 to match $1 - \alpha = 0.95$ exactly.

Pivotal quantity. The key for constructing a random interval is to find a **pivotal quantity** (pivot) whose distribution does not depend on the unknown parameter, such as $Z = \dfrac{\overline{X} - \mu_X}{\sigma_X/\sqrt{n}} \sim N(0,1)$. Once the pivot and its distribution are known, its quantiles determine the endpoints.

Why equal tails? The choice is not unique: for instance $\mathbb{P}\big(\overline{X} - z_{\alpha/4}\dfrac{\sigma_X}{\sqrt{n}} \leq \mu_X \leq \overline{X} + z_{3\alpha/4}\dfrac{\sigma_X}{\sqrt{n}}\big) = 1 - \alpha$ also holds, but this interval is longer than the symmetric one at

the same confidence level, i.e. it has lower precision. The equal-tail (symmetric) interval is the shortest.

4.3.3 Two Sampling Distributions

To construct confidence intervals we need the sampling distribution of the point estimator. Two distributions play a central role when the population is normal.

χ^2 distribution. Let $X_1, X_2, \ldots, X_n$ be independent with $X_i \sim N(0,1)$, and let $Y = X_1^2 + X_2^2 + \cdots + X_n^2$. Then Y follows the χ^2 **distribution with n degrees of freedom**, denoted $Y \sim \chi^2(n)$, whose pdf is $f(y) = \dfrac{1}{2^{n/2}\Gamma(n/2)}\, y^{n/2-1} e^{-y/2}$ for $y > 0$, and $f(y) = 0$ for $y \le 0$.

Degrees of freedom. The degrees of freedom are the number of variables that can change independently.

Additivity. If $V \sim \chi^2(n_1)$, $W \sim \chi^2(n_2)$ and V, W are independent, then $V + W \sim \chi^2(n_1 + n_2)$.

Proof: write $V = X_1^2 + X_2^2 + \cdots + X_{n_1}^2$ where $X_1, \ldots, X_{n_1}$ are i.i.d. $N(0,1)$, and $W = Z_1^2 + Z_2^2 + \cdots + Z_{n_2}^2$ where $Z_1, \ldots, Z_{n_2}$ are i.i.d. $N(0,1)$. Then $V + W = X_1^2 + \cdots + X_{n_1}^2 + Z_1^2 + \cdots + Z_{n_2}^2$ is a sum of $n_1 + n_2$ independent squared standard normals, hence $V + W \sim \chi^2(n_1 + n_2)$. ■

Numerical description. If $Y \sim \chi^2(n)$, then $\boxed{\mathbb{E}(Y) = n}$ and $\boxed{\mathrm{Var}(Y) = 2n}$.

Proof: **Mean.** Y and $X_1^2 + \cdots + X_n^2$ have the same distribution with X_i i.i.d. $N(0,1)$, so $\mathbb{E}(Y) = \sum_{i=1}^{n} \mathbb{E}(X_i^2) = \sum_{i=1}^{n} 1 = n$, using $\mathbb{E}(X_i^2) = \mathrm{Var}(X_i) = 1$.

Variance. By independence, $\mathrm{Var}(Y) = \sum_{i=1}^{n} \mathrm{Var}(X_i^2) = n\,\mathrm{Var}(X_1^2)$, and $\mathrm{Var}(X_1^2) = \mathbb{E}(X_1^4) - [\mathbb{E}(X_1^2)]^2 = \displaystyle\int_{-\infty}^{\infty} x^4 \frac{1}{\sqrt{2\pi}} e^{-x^2/2}\, dx - 1 = 3 - 1 = 2$, where $\mathbb{E}(X_1^4) = 3$ follows from integration by parts. Hence $\mathrm{Var}(Y) = 2n$. ■

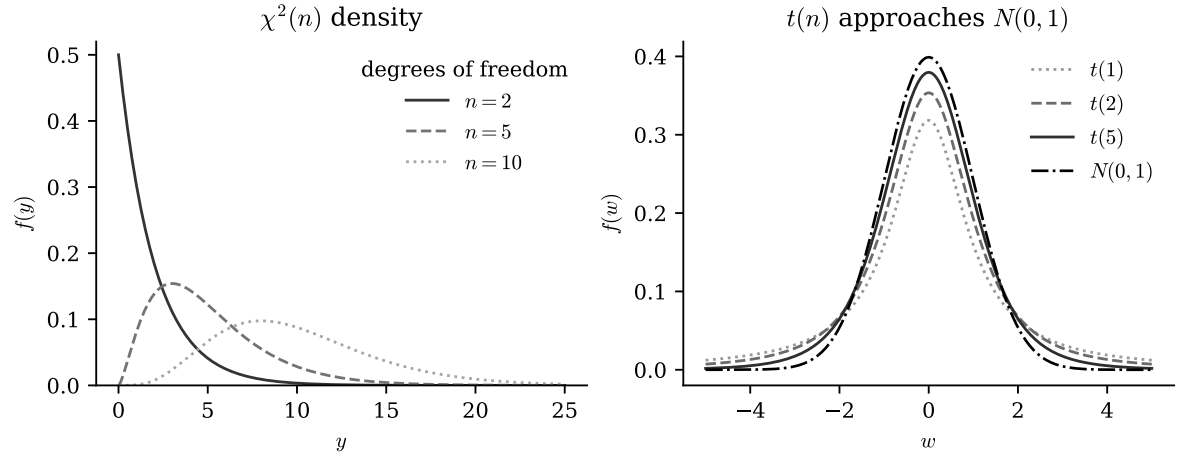
t distribution (Student). Let $Z \sim N(0,1)$ and $Y \sim \chi^2(n)$ be independent. Then $T = \dfrac{Z}{\sqrt{Y/n}} \sim t(n)$, with pdf $f(w) = \dfrac{\Gamma((n+1)/2)}{\sqrt{n\pi}\,\Gamma(n/2)} \left(1 + \dfrac{w^2}{n}\right)^{-(n+1)/2}$ for $-\infty < w < \infty$.
Properties:

- $f(-w) = f(w)$; $f'(w) > 0$ for $w < 0$ and $f'(w) < 0$ for $w > 0$; $f(w) \to 0$ as $w \to \pm\infty$. The curve is bell-shaped with a peak at 0 and has heavier tails than $N(0,1)$.
- $\mathbb{E}(T) = 0$ for $n > 1$, $\mathrm{Var}(T) = \dfrac{n}{n-2}$ for $n > 2$.
- As $n \to \infty$, $t(n) \to N(0,1)$; the density curves for $k = 1, 2, 5, +\infty$ approach the standard normal curve.

R corner: use `qt(p, df)` for t quantiles, e.g. $t_{3,0.05}$ is `qt(0.05, df = 3, lower.tail = FALSE)`; use `qchisq(p, df)` for χ^2 quantiles.

The t distribution was introduced by William Sealy Gosset (1908, *Biometrika*), who

published under the pseudonym "Student"; hence the name Student's t distribution.



4.3.4 Three Sampling Distribution Theorems

Throughout 4.3 we assume a normal population: $X \sim N(\mu_X, \sigma_X^2)$.

Theorem 1: $\boxed{\overline{X} \sim N\left(\mu_X, \sigma_X^2/n\right)}$, equivalently $\boxed{\dfrac{\overline{X}-\mu_X}{\sigma_X/\sqrt{n}} \sim N(0,1)}$.

Lemma (linear combinations of normals): if $X \sim N(\mu_1, \sigma_1^2)$ and $Y \sim N(\mu_2, \sigma_2^2)$ are independent, then for any constants a, b, $aX+bY \sim N(a\mu_1+b\mu_2, a^2\sigma_1^2+b^2\sigma_2^2)$; by iteration the same holds for any finite linear combination of independent normal r.v.'s.

Proof of Theorem 1: $X_1, X_2, \ldots, X_n$ are i.i.d. $N(\mu_X, \sigma_X^2)$, and $\overline{X} = \dfrac{1}{n}X_1 + \dfrac{1}{n}X_2 + \cdots + \dfrac{1}{n}X_n$ is a linear combination with $a_i = 1/n$. By the lemma, $\overline{X}$ is normal. For the parameters, $\mathbb{E}(\overline{X}) = \displaystyle\sum_{i=1}^{n} \frac{1}{n}\mathbb{E}(X_i) = \frac{1}{n} \cdot n\mu_X = \mu_X$, and by independence $\operatorname{Var}(\overline{X}) = \displaystyle\sum_{i=1}^{n} \frac{1}{n^2}\operatorname{Var}(X_i) = \frac{1}{n^2} \cdot n\sigma_X^2 = \frac{\sigma_X^2}{n}$. Hence $\overline{X} \sim N\left(\mu_X, \dfrac{\sigma_X^2}{n}\right)$, and standardizing gives $\dfrac{\overline{X}-\mu_X}{\sigma_X/\sqrt{n}} \sim N(0,1)$. ■

Theorem 2: $\boxed{\dfrac{(n-1)S_{n-1}^2}{\sigma_X^2} \sim \chi^2(n-1)}$, and $\overline{X}$ and S_{n-1}^2 are *independent*.

Proof of the algebraic identity. Write $X_i - \overline{X} = (X_i - \mu_X) - (\overline{X} - \mu_X)$ and expand the square:

$$\begin{aligned}(n-1)S_{n-1}^2 &= \sum_{i=1}^{n}(X_i - \overline{X})^2 = \sum_{i=1}^{n}\big[(X_i-\mu_X) - (\overline{X}-\mu_X)\big]^2 \\ &= \sum_{i=1}^{n}(X_i-\mu_X)^2 - 2(\overline{X}-\mu_X)\sum_{i=1}^{n}(X_i-\mu_X) + n(\overline{X}-\mu_X)^2.\end{aligned}$$

Since $\sum_{i=1}^{n}(X_i-\mu_X)=n(\overline{X}-\mu_X)$, the middle term is $-2n(\overline{X}-\mu_X)^2$, so $(n-1)S_{n-1}^2=\sum_{i=1}^{n}(X_i-\mu_X)^2-n(\overline{X}-\mu_X)^2$. Dividing by σ_X^2 and completing the squares, $\dfrac{(n-1)S_{n-1}^2}{\sigma_X^2}=\sum_{i=1}^{n}\left(\dfrac{X_i-\mu_X}{\sigma_X}\right)^2-\left(\dfrac{\overline{X}-\mu_X}{\sigma_X/\sqrt{n}}\right)^2$.

Why $\chi^2(n-1)$. The first term is a sum of n squares of independent $N(0,1)$ variables, hence $\chi^2(n)$; the second term is a single squared $N(0,1)$ variable, hence $\chi^2(1)$.

Because $\overline{X}$ enters both terms they are *not* independent, so naively subtracting does not give a χ^2 distribution.

A further orthogonal change of variables (Cochran's theorem) shows that $\dfrac{(n-1)S_{n-1}^2}{\sigma_X^2}$ can be written as a sum of exactly $n-1$ independent squared standard normals, and that this sum is independent of $\overline{X}$. Hence $\dfrac{(n-1)S_{n-1}^2}{\sigma_X^2}\sim\chi^2(n-1)$ and $\overline{X}$ and S_{n-1}^2 are independent. ■

Remark (why $n-1$, and the variability of S_{n-1}^2). Since $(n-1)S_{n-1}^2/\sigma_X^2\sim\chi^2(n-1)$, its mean is $n-1$ and its variance is $2(n-1)$.

Taking expectations, $\dfrac{n-1}{\sigma_X^2}\mathbb{E}(S_{n-1}^2)=n-1$, hence $\boxed{\mathbb{E}(S_{n-1}^2)=\sigma_X^2}$: the estimator is unbiased.

For the variance, $\mathrm{Var}(S_{n-1}^2)=\dfrac{\sigma_X^4}{(n-1)^2}\mathrm{Var}\big((n-1)S_{n-1}^2/\sigma_X^2\big)=\dfrac{\sigma_X^4}{(n-1)^2}\cdot 2(n-1)=\boxed{2\sigma_X^4/(n-1)}$.

This also indicates why S_{n-1}^2 is used to replace σ_X^2 when σ_X^2 is unknown.

Theorem 3: $\boxed{\dfrac{\overline{X}-\mu_X}{S_{n-1}/\sqrt{n}}\sim t(n-1)}$.

Proof: By Theorem 1, $U=\dfrac{\overline{X}-\mu_X}{\sigma_X/\sqrt{n}}\sim N(0,1)$. By Theorem 2, $V=\dfrac{(n-1)S_{n-1}^2}{\sigma_X^2}\sim\chi^2(n-1)$, and U and V are independent. By the definition of the t distribution,

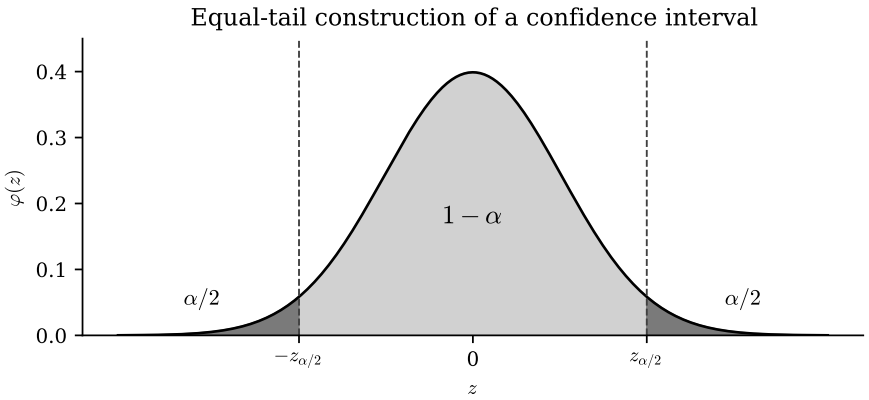
$$\frac{U}{\sqrt{V/(n-1)}}=\frac{\frac{\overline{X}-\mu_X}{\sigma_X/\sqrt{n}}}{\sqrt{\frac{(n-1)S_{n-1}^2}{\sigma_X^2(n-1)}}}=\frac{\overline{X}-\mu_X}{S_{n-1}/\sqrt{n}}\sim t(n-1).\qquad\blacksquare$$

4.3.5 Confidence Interval for μ_X: σ_X^2 Known

If σ_X^2 is known, by Theorem 1 the pivot is $Z=\dfrac{\overline{X}-\mu_X}{\sigma_X/\sqrt{n}}\sim N(0,1)$. Notation: for $Z\sim N(0,1)$, z_a denotes its **upper a-th quantile**, i.e. $\mathbb{P}(Z>z_a)=a$ for $a\in(0,1)$; by

symmetry $\mathbb{P}(Z < -z_a) = a$ and $\mathbb{P}(-z_a \le Z \le z_a) = 1 - 2a$.

Split the error probability equally between the two tails, $\alpha/2$ in each: $\mathbb{P}\big(-z_{\alpha/2} \le \dfrac{\overline{X} - \mu_X}{\sigma_X/\sqrt{n}} \le z_{\alpha/2}\big) = 1 - \alpha$.



Now isolate μ_X in three reversible steps.

1. Multiply all parts by $\sigma_X/\sqrt{n} > 0$: $\mathbb{P}\big(-z_{\alpha/2}\dfrac{\sigma_X}{\sqrt{n}} \le \overline{X} - \mu_X \le z_{\alpha/2}\dfrac{\sigma_X}{\sqrt{n}}\big) = 1 - \alpha$.
2. Subtract $\overline{X}$ from all parts: $\mathbb{P}\big(-\overline{X} - z_{\alpha/2}\dfrac{\sigma_X}{\sqrt{n}} \le -\mu_X \le -\overline{X} + z_{\alpha/2}\dfrac{\sigma_X}{\sqrt{n}}\big) = 1 - \alpha$.
3. Multiply by -1, which reverses the inequalities: $\mathbb{P}\big(\overline{X} - z_{\alpha/2}\dfrac{\sigma_X}{\sqrt{n}} \le \mu_X \le \overline{X} + z_{\alpha/2}\dfrac{\sigma_X}{\sqrt{n}}\big) = 1 - \alpha$.

So the random interval inside the last probability contains μ_X with probability $1-\alpha$, and substituting the observed $\overline{x}$ gives the confidence interval

Confidence interval for μ_X (σ_X^2 known)

$\overline{x} \pm z_{\alpha/2} \cdot \dfrac{\sigma_X}{\sqrt{n}}$ For a 95% C.I., $z_{0.025} = 1.96$, so the interval is $\overline{x} \pm 1.96 \cdot \dfrac{\sigma_X}{\sqrt{n}}$.

CI for μ_X with known σ_X^2

A random sample of $n = 20$ from $N(\mu_X, \sigma_X^2)$ with $\sigma_X^2 = 225$ ($\sigma_X = 15$) has sample mean $\overline{x} = 64.3$. Construct a 95% confidence interval for μ_X. $\overline{x} \pm z_{0.025} \cdot \dfrac{\sigma_X}{\sqrt{n}} = 64.3 \pm 1.96 \cdot \dfrac{15}{\sqrt{20}} = 64.3 \pm 6.57 = [57.73, 70.87]$. We are 95% confident that the true population mean lies between 57.7 and 70.9.

Interpretation. The numerical interval $[57.73, 70.87]$ is not random, so the probability that it contains μ_X is either 0 (it never does) or 1 (it certainly does). The value 0.95 refers to the procedure: with B different sets of data we obtain B intervals $\overline{x} \pm 1.96\,\sigma_X/\sqrt{n}$, some containing μ_X and some not, and for large B the proportion containing μ_X tends to 0.95.

4.3.6 Confidence Interval for μ_X: σ_X^2 Unknown

When σ_X^2 is unknown, we replace it with S_{n-1}^2. By Theorem 3 the pivot is now $T_{n-1} = \dfrac{\overline{X} - \mu_X}{S_{n-1}/\sqrt{n}} \sim t(n-1)$; note that the denominator S_{n-1} is itself a random variable, not a number, which is exactly why the t distribution replaces $N(0,1)$.

Let $t_{n-1,a}$ denote the upper a-th quantile of $t(n-1)$, i.e. $\mathbb{P}(T_{n-1} > t_{n-1,a}) = a$; by symmetry of the t density, $\mathbb{P}(T_{n-1} < -t_{n-1,a}) = a$. With $\alpha/2$ in each tail, $\mathbb{P}\left(-t_{n-1,\alpha/2} \le \dfrac{\overline{X} - \mu_X}{S_{n-1}/\sqrt{n}} \le t_{n-1,\alpha/2}\right) = 1-\alpha$.

Multiplying by $S_{n-1}/\sqrt{n} > 0$, subtracting $\overline{X}$ and multiplying by -1 (the same three reversible steps as above) gives $\mathbb{P}\left(\overline{X} - t_{n-1,\alpha/2}\dfrac{S_{n-1}}{\sqrt{n}} \le \mu_X \le \overline{X} + t_{n-1,\alpha/2}\dfrac{S_{n-1}}{\sqrt{n}}\right) = 1-\alpha$, so the observed confidence interval is

Confidence interval for μ_X (σ_X^2 unknown)

$\overline{x} \pm t_{n-1,\alpha/2} \cdot \dfrac{s_{n-1}}{\sqrt{n}}$ Use `qt(1 - α/2, df = n-1)` in R.

CI for μ_X with unknown σ_X^2

A paint manufacturer tested 12 areas and obtained a mean drying time of 66.3 minutes with sd 8.4 minutes. Assume drying time is normal. Construct a 95% C.I. for the population mean.
We have $n = 12$, $\overline{x} = 66.3$, $s_{n-1} = 8.4$, $t_{11,0.025} = 2.201$ (from R: `qt(0.975, 11)`).
So $66.3 \pm 2.201 \cdot \dfrac{8.4}{\sqrt{12}} = 66.3 \pm 5.32 = [60.98, 71.62]$.

```
# R corner: t.test for elephant call frequencies
freq <- c(14, 16, 17, 17, 24, 20, 32, 18, 29, 31, 15, 35)
t.test(freq, conf.level = 0.95)   # 95% CI for mu (sigma^2 unknown)
# qt(0.975, df = 11) gives 2.200985
```

Increasing the confidence level to 99% uses `conf.level = 0.99`, i.e. $t_{11,0.005} = 3.106$ instead of $t_{11,0.025} = 2.201$, and widens the interval to $22.33 \pm 3.106 \cdot 7.51/\sqrt{12} \approx [15.60, 29.07]$: higher confidence is paid for by lower precision.

4.3.7 Confidence Interval for σ_X^2

When μ_X is unknown, by Theorem 2 the pivot is $U_{n-1} = \dfrac{(n-1)S_{n-1}^2}{\sigma_X^2} \sim \chi^2(n-1)$, with the sample variance in the numerator being a random variable. Notation: $\chi^2_{n-1,a}$ denotes the upper a-th quantile of $\chi^2(n-1)$, i.e. $\mathbb{P}(U_{n-1} > \chi^2_{n-1,a}) = a$. Splitting the error probability as $\alpha/2$ in each tail, $\mathbb{P}\left(\chi^2_{n-1,1-\alpha/2} \le \dfrac{(n-1)S_{n-1}^2}{\sigma_X^2} \le \chi^2_{n-1,\alpha/2}\right) = 1-\alpha$. Since all three parts are positive, taking reciprocals reverses the inequalities:

$\mathbb{P}\left(\dfrac{1}{\chi^2_{n-1,1-\alpha/2}} \geq \dfrac{\sigma_X^2}{(n-1)S_{n-1}^2} \geq \dfrac{1}{\chi^2_{n-1,\alpha/2}}\right) = 1-\alpha$; multiplying by $(n-1)S_{n-1}^2 > 0$ then gives $\mathbb{P}\left(\dfrac{(n-1)S_{n-1}^2}{\chi^2_{n-1,\alpha/2}} \leq \sigma_X^2 \leq \dfrac{(n-1)S_{n-1}^2}{\chi^2_{n-1,1-\alpha/2}}\right) = 1-\alpha$, so the observed confidence interval for σ_X^2 is

Confidence interval for σ_X^2

$$\left[\frac{(n-1)s_{n-1}^2}{\chi^2_{n-1,\alpha/2}}, \frac{(n-1)s_{n-1}^2}{\chi^2_{n-1,1-\alpha/2}}\right]$$

The χ^2 distribution is asymmetric, so the interval is not symmetric about the point estimate s_{n-1}^2.

```
# R corner: chi-squared quantiles for n = 10
qchisq(0.975, df = 10)              # chi2_{10,0.975} = 3.247
qchisq(0.025, df = 10)              # chi2_{10,0.025} = 20.483
# VarCI from DescTools, elephant data
# VarCI(freq, conf.level = 0.95)
```

For the elephant data $n = 12$ and $s_{n-1}^2 \approx 56.42$ with $\chi^2_{11,0.975} = 3.816$ and $\chi^2_{11,0.025} = 21.920$, so the 95% C.I. for σ_X^2 is $\left[\dfrac{11 \cdot 56.42}{21.920}, \dfrac{11 \cdot 56.42}{3.816}\right] \approx [28.3, 162.6]$, which agrees with `VarCI`.

Exercise (μ_X known). If μ_X is known, Theorem 2 is replaced by the pivot $\sum_{i=1}^{n}(X_i - \mu_X)^2/\sigma_X^2 \sim \chi^2(n)$ (see the summary table below), so the limits use n instead of $n-1$ and $\sum_{i=1}^{n}(x_i - \mu_X)^2$ instead of $(n-1)s_{n-1}^2$.

4.3.8 Summary: Pivotal Quantities

Parameter	Other parameter	Pivotal quantity and its distribution
μ_X	σ_X^2 known	$\dfrac{\overline{X} - \mu_X}{\sigma_X/\sqrt{n}} \sim N(0,1)$
μ_X	σ_X^2 unknown	$\dfrac{\overline{X} - \mu_X}{S_{n-1}/\sqrt{n}} \sim t(n-1)$
σ_X^2	μ_X known	$\displaystyle\sum_{i=1}^{n}\left(\frac{X_i - \mu_X}{\sigma_X}\right)^2 \sim \chi^2(n)$
σ_X^2	μ_X unknown	$\dfrac{(n-1)S_{n-1}^2}{\sigma_X^2} \sim \chi^2(n-1)$

4.3.9 Central Limit Theorem (CLT, optional remark)

So far we have assumed the population is normal. When it is not, exact sampling distributions of $\overline{X}$ are generally intractable. The **Central Limit Theorem** gives a large-n approximation:

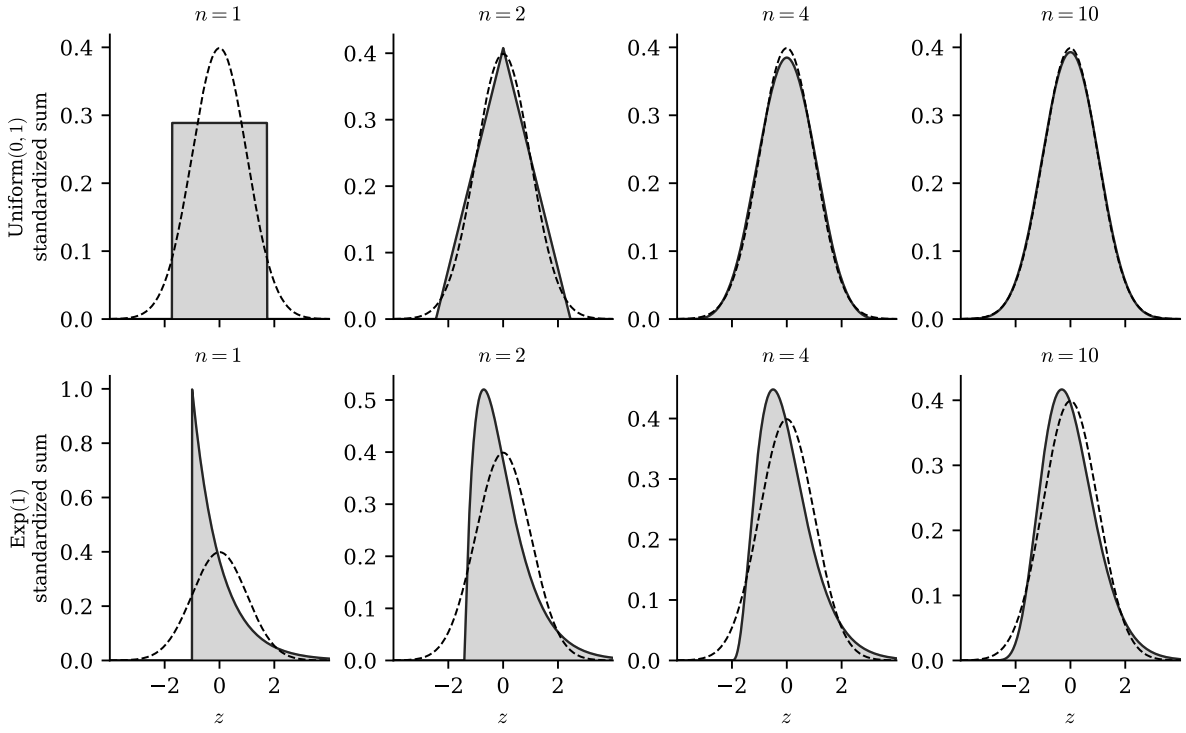
Theorem (CLT): If $\overline{X}$ is the mean of a random sample of size n from *any* population with mean μ_X and finite variance σ_X^2, then as $n \to \infty$, $\boxed{\dfrac{\overline{X} - \mu_X}{\sigma_X/\sqrt{n}} \xrightarrow{d} N(0,1)}$.

The two standardizations in the statement are the same: since $\displaystyle\sum_{i=1}^{n} X_i = n\overline{X}$, $Z = \dfrac{\sum_{i=1}^{n} X_i - n\mu_X}{\sqrt{n}\,\sigma_X} = \dfrac{n\overline{X} - n\mu_X}{\sqrt{n}\,\sigma_X} = \dfrac{\overline{X} - \mu_X}{\sigma_X/\sqrt{n}}$; the numerator has mean 0 and the denominator is its standard deviation, so Z is the standardized sum.

The symbol $\xrightarrow{d}$ means *converges in distribution*. A numerical illustration shows the mechanism: if $X_i \overset{iid}{\sim}$ Uniform$(0,1)$ (density flat on $[0,1]$) or $X_i \overset{iid}{\sim}$ Exp(1) (density e^{-x} on $x \geq 0$), then the density of $Y_n = \displaystyle\sum_{i=1}^{n} X_i$ is flat (respectively exponential) for $n = 1$, becomes triangular for $n = 2$, and turns bell-shaped as n grows; after standardization it approaches the $N(0,1)$ density.

Two practical consequences:

- **Rule of thumb:** the normal approximation is reasonable for $n \geq 30$, and works surprisingly well even for skewed distributions when n is large.
- **Why normality is everywhere:** many real-world quantities are sums of many small independent effects, hence approximately normal; this is also why the sampling distribution of $\overline{X}$ is central in statistical inference.



Dashed curve: $N(0,1)$; shaded curve: standardized density of $X_1 + \cdots + X_n$

■

5 Hypothesis Testing

5.1 Concepts in Hypothesis Testing

5.1.1 Motivation: Statistical Inference

Statistical inference is the process of drawing conclusions about populations, or about scientific truths, from the collected data. It answers questions such as whether a manufacturing process is acceptable, whether a new drug is effective, and so on.

In Chapter 4 we used a sample to obtain a point-valued or interval-valued *estimate* of an unknown parameter. Here the logic is reversed: we *assume* a hypothesized value of the parameter and then use the data to decide whether that assumption should be rejected.

Example (quality control): A manufacturing process is sampled: 100 items are inspected and 10 are found to be defective. In the long run the company can tolerate at most 5% defective items. Is the process acceptable?

Analysis: assume that the process is acceptable, i.e., the long-run defective rate does not exceed 5%. Then the number of defective items $DI \sim \text{Binomial}(100, 0.05)$, and

$$\mathbb{P}(DI \geq 10) = 1 - \mathbb{P}(DI \leq 9) = 1 - \sum_{k=0}^{9} \binom{100}{k} (0.05)^k (0.95)^{100-k} \approx 0.0282.$$

This probability is very small, so obtaining 10 or more defective items is a rare event under the assumption; the data therefore suggest that the long-run defective rate exceeds 5%. If instead only 6 (or 4) items were defective, the evidence would be much weaker and the conclusion far less clear-cut.

5.1.2 Null and Alternative Hypotheses

Definition: a **statistical hypothesis** is an assertion, conjecture or statement about the population, usually formulated in terms of population parameters.

In this chapter the statements concern the parameter μ_X or σ_X^2, and we always have a pair:

- **Null hypothesis H_0:** the hypothesis that is assumed to be true and then tested to be rejected or not rejected. It always contains an equality sign ($=$, $\leq$ or $\geq$).
- **Alternative hypothesis H_1:** the hypothesis that contains the values of the parameter we would accept if H_0 is rejected. It never contains an equality sign, except in a simple test.

Example: An agronomist wants to decide whether a new fertilizer gives a higher mean yield than the old one, whose mean is known to be 10. Assuming normality, the hypotheses are $H_0 : \mu_X = 10$ against $H_1 : \mu_X > 10$. A pharmaceutical manufacturer wants to decide whether 90% of all patients given a new medication recover, i.e., whether the binomial parameter p equals 0.90: $H_0 : p = 0.9$ against $H_1 : p \neq 0.9$.

5.1.3 Types of Hypothesis Testing

According to the form of the alternative hypothesis, tests on μ_X are classified as follows.

Type	H_0	H_1
Simple test	$\mu_X = \mu_0$	$\mu_X = \mu_1$
One-sided right test	$\mu_X = \mu_0$	$\mu_X > \mu_0$
One-sided left test	$\mu_X = \mu_0$	$\mu_X < \mu_0$
Two-sided test	$\mu_X = \mu_0$	$\mu_X \neq \mu_0$

In this course we do *not* study any test with an inequality null hypothesis (details belong to MATH 3423). Unless there is enough information to justify a simple or one-sided test, the default choice in practice is the two-sided test.

5.1.4 Basic Idea of Hypothesis Testing

The basic idea is a **contradiction**, carried out in three steps:

1. Determine H_0 and H_1.
2. Under H_0, define a **rare event** — an event that happens with a very small probability in one experiment of getting n data.
3. Collect the data. If the data contradict H_0, then H_0 is false and we *reject* H_0; if the data do not contradict H_0, we cannot say that H_0 is true, but we say that we *do not reject* H_0.

Example (fair coin): We want to know whether a coin is fair, and we flip it 10 times.

Step 1: H_0: the coin is fair, i.e., $\mathbb{P}(H) = \mathbb{P}(T) = 1/2$; H_1: the coin is not fair.

Step 2: under H_0, the probability of getting 10 tails in one experiment is $(1/2)^{10} \approx 0.00098$, so "10 tails" is a rare event under H_0.

Step 3: if the experiment does produce 10 tails, the data contradict H_0; we then have evidence to suspect the reliability of H_0 and hence reject H_0 (and accept H_1).

Remark: In hypothesis testing we only use the data to see whether there is enough evidence to reject H_0. If there is, we can be confident that H_0 is false and H_1 is true. If there is not, it does *not* mean that we are confident that H_0 is true; we should say "do not reject H_0" instead of "accept H_0".

5.1.5 Test Errors and Error Probabilities

A **test statistic** (a point-valued estimator) is used to formulate a **test statement** of H_0: a procedure based on the observed values of the random variable of interest that leads to the rejection or non-rejection of H_0. No test statement is perfect; each one leads to the following two kinds of errors.

	Do not reject H_0	Reject H_0
H_0 is true	No error	**Type I error**
H_0 is false	**Type II error**	No error

Type I error: the error of rejecting H_0 when it is in fact true.

Type II error: the error of not rejecting H_0 when it is in fact false.

An analogous medical example: let H_0 be "the patient is not infected" and let "reject H_0" be "tested positive". Then a false positive (testing positive although healthy) is a Type I error, and a false negative (testing negative although infected) is a Type II error.

The error probabilities are therefore

Error probabilities

$$\alpha = \mathbb{P}(\text{Type I error}) = \mathbb{P}(\text{reject } H_0 \mid H_0 \text{ is true}),$$
$$\beta = \mathbb{P}(\text{Type II error}) = \mathbb{P}(\text{not reject } H_0 \mid H_0 \text{ is false}).$$

Ideally both probabilities would be small, but for a fixed sample size they cannot be controlled simultaneously. In designing a test we normally guarantee α at a desired low value (often 0.01, 0.05 or 0.1) and then make β as small as possible. There is a trade-off between the two errors: making α smaller makes β larger, and vice versa.

Example (light bulbs): Light bulbs from a standard process have lifetimes distributed as $N(\mu_X, 300^2)$. The standard deviation $\sigma_X = 300$ is known, but the mean μ_X is not; for simplicity we are sure that μ_X is either 1200 or 1240. Consider the simple test $H_0 : \mu_X = 1200$ against $H_1 : \mu_X = 1240$.

A random sample of $n = 100$ bulbs is drawn and $\overline{X}$ estimates μ_X. Since the hypothesized value under H_1 is larger, a large $\overline{x}$ should lead to rejection, so we use the test statement "reject H_0 if $\overline{x} > c$", where c is called a **critical value**. Then $\alpha = \mathbb{P}(\overline{X} > c \mid \mu_X = 1200)$ and $\beta = \mathbb{P}(\overline{X} \le c \mid \mu_X = 1240)$.

Under H_0, $\overline{X} \sim N(1200, 300^2/100)$, so $\{\overline{X} > c\}$ is a rare event under H_0; observing $\overline{x} > c$ in one experiment contradicts H_0 and hence leads to rejection.

5.1.6 Determination of the Critical Value

We test $H_0 : \mu_X = 1200$ against $H_1 : \mu_X = 1240$ at the significance level $\alpha = 0.05$ using "reject H_0 if $\overline{x} > c$". The critical value is fixed by the Type I error probability:

$$\mathbb{P}(\overline{X} > c \mid \mu_X = 1200) = \alpha \implies \frac{c - 1200}{300/\sqrt{100}} = z_{0.05} \implies \boxed{c = 1200 + 30\, z_{0.05} = 1249.35}.$$

The test statement is therefore: *reject H_0 at significance level* 0.05 *if* $\overline{x} > 1249.35$. The set $\{x_1, \ldots, x_n : \overline{x} > 1249.35\}$ is the **rejection region** (or critical region), and $\overline{X}$ is the test statistic.

Discussion: if $\overline{x} = 1301$, we reject H_0 at level $\alpha = 0.05$; if $\overline{x} = 1237$, we do not have enough evidence to reject H_0 at level $\alpha = 0.05$.

5.1.7 p-value

Roughly speaking, the **p-value** is the probability of getting our data under H_0. By the statistical belief of Section 5.1.4, if this probability is small then H_0 is likely wrong. "Small" is judged by comparing the p-value with the significance level α:

- if p-value $< \alpha$, there is enough evidence to reject H_0 at level α;
- if p-value $> \alpha$, there is not enough evidence to reject H_0 at that level.

For the light-bulb example, with observed mean $\overline{x}$ and $Z \sim N(0,1)$, p-value $= \mathbb{P}(\overline{X} > \overline{x} \mid \mu_X = 1200) = \mathbb{P}\left(Z > \dfrac{\overline{x} - 1200}{30}\right)$.

Thus $\overline{x} = 1301$ gives p-value $= 1 - \Phi(3.3667) \approx 0.0004 < 0.05$ (reject), while $\overline{x} = 1237$ gives p-value $= 1 - \Phi(1.2333) \approx 0.109 > 0.05$ (do not reject).

Equivalence of the two approaches. For any test the two statements coincide:

$$\boxed{\text{reject } H_0 \text{ at level } \alpha \iff \text{p-value} < \alpha}.$$

That is, "the observed data fall in the rejection region" and "the p-value is below α" say the same thing. Hence there are two ways to make the decision:

- **Method 1 (critical value):** state H_0 and H_1; choose a fixed significance level α; choose a test statistic and establish the critical region; reject H_0 if the computed statistic lies in the critical region, otherwise do not reject; draw the conclusion.
- **Method 2 (p-value):** state H_0 and H_1; choose a test statistic; compute the p-value from the observed statistic; use judgment based on the p-value and knowledge of the scientific system.

5.1.8 Power of a Test Statement

The **power** of a test statement measures the goodness of the test:

Power of a test

power $= 1 - \beta = \mathbb{P}(\text{reject } H_0 \mid H_0 \text{ is false})$. Two test statements can be compared only when they share the same α; then the one with the larger power is better.

For the test "reject H_0 at level 0.05 if $\overline{x} > 1249.35$" of the light-bulb example, $\beta = \mathbb{P}(\overline{X} \le 1249.35 \mid \mu_X = 1240) = \Phi\left(\dfrac{1249.35 - 1240}{30}\right) = \Phi(0.3117) \approx 0.6224$, so the power is $\boxed{1 - \beta = 0.3776}$.

If the sample size is increased to $n = 400$, then $\text{sd}(\overline{X}) = 300/\sqrt{400} = 15$ and $\dfrac{c - 1200}{15} = z_{0.05} = 1.645 \implies c = 1224.675$. The test statement becomes "reject H_0 at level 0.05 if $\overline{x} > 1224.675$", and $\beta = \mathbb{P}(\overline{X} \le 1224.675 \mid \mu_X = 1240) = \Phi(-1.0217) \approx 0.1535$, so the power rises to $\boxed{1 - \beta = 0.8465}$.

Two lessons: using more data increases the power of the test without changing α; conversely, at a fixed sample size, increasing the power requires accepting a larger α.

5.2 Hypothesis Testing: Normal Case

Theorem (sampling distributions, Section 4.3): Let $X_1, X_2, \ldots, X_n$ be a random sample from $X \sim N(\mu_X, \sigma_X^2)$. Then

- $\boxed{\overline{X} \sim N\left(\mu_X, \sigma_X^2/n\right)}$,
- $\boxed{(n-1)S_{n-1}^2/\sigma_X^2 \sim \chi^2(n-1)}$,
- $\boxed{(\overline{X} - \mu_X)/(S_{n-1}/\sqrt{n}) \sim t(n-1)}$,

and $\overline{X}$ and S_{n-1}^2 are independent.

From now on we only consider one-sided and two-sided tests when $X \sim N(\mu_X, \sigma_X^2)$, first with σ_X^2 known and then with σ_X^2 unknown.

5.2.1 Tests for μ_X: σ_X^2 Known

One-sided right test. Consider $H_0 : \mu_X = \mu_0$ against $H_1 : \mu_X > \mu_0$. A large $\overline{x}$ is evidence against H_0, so the test statement is "reject H_0 if $\overline{x} > c$".

Under H_0, $\overline{X} \sim N(\mu_0, \sigma_X^2/n)$. With $Z = \dfrac{\overline{X} - \mu_0}{\sigma_X/\sqrt{n}} \sim N(0,1)$, $\alpha = \mathbb{P}(\overline{X} > c \mid \mu_X = \mu_0) = \mathbb{P}\left(Z > \dfrac{c-\mu_0}{\sigma_X/\sqrt{n}}\right)$, hence $\dfrac{c-\mu_0}{\sigma_X/\sqrt{n}} = z_\alpha$, i.e. $\boxed{c = \mu_0 + z_\alpha \dfrac{\sigma_X}{\sqrt{n}}}$.

So we reject H_0 iff $\overline{x} > \mu_0 + z_\alpha \dfrac{\sigma_X}{\sqrt{n}}$, i.e. iff the observed $z = \dfrac{\overline{x} - \mu_0}{\sigma_X/\sqrt{n}} > z_\alpha$. The p-value is the probability, under H_0, of a result at least as extreme as the observed one: p-value $= \mathbb{P}(\overline{X} > \overline{x} \mid \mu_X = \mu_0) = \mathbb{P}\left(Z > \dfrac{\overline{x}-\mu_0}{\sigma_X/\sqrt{n}}\right) = \mathbb{P}(Z > z)$.

One-sided left test. For $H_0 : \mu_X = \mu_0$ against $H_1 : \mu_X < \mu_0$ we reject H_0 if $\overline{x} < c$. Then $\alpha = \mathbb{P}(\overline{X} < c \mid \mu_X = \mu_0) = \mathbb{P}\left(Z < \dfrac{c-\mu_0}{\sigma_X/\sqrt{n}}\right)$, hence $\dfrac{c-\mu_0}{\sigma_X/\sqrt{n}} = -z_\alpha$, i.e. $\boxed{c = \mu_0 - z_\alpha \dfrac{\sigma_X}{\sqrt{n}}}$.

So we reject H_0 iff $\overline{x} < \mu_0 - z_\alpha \dfrac{\sigma_X}{\sqrt{n}}$, i.e. iff $z < -z_\alpha$, with p-value $\mathbb{P}(Z < z)$.

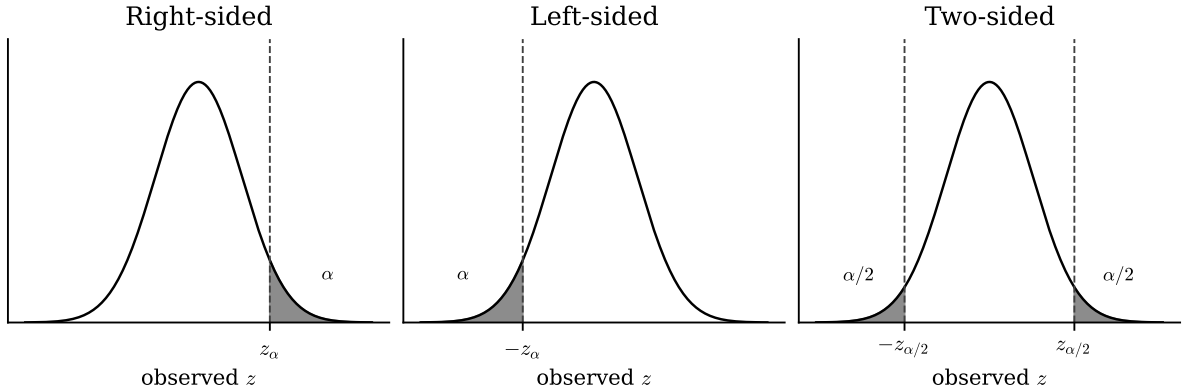
Two-sided test. For $H_0 : \mu_X = \mu_0$ against $H_1 : \mu_X \neq \mu_0$ we reject H_0 if $\overline{x} < a$ or $\overline{x} > b$. Splitting the Type I error equally, $\alpha/2$ in each tail, $\mathbb{P}(\overline{X} < a \mid \mu_X = \mu_0) + \mathbb{P}(\overline{X} > b \mid \mu_X = \mu_0) = \dfrac{\alpha}{2} + \dfrac{\alpha}{2} = \alpha$, which gives $\boxed{a = \mu_0 - z_{\alpha/2}\dfrac{\sigma_X}{\sqrt{n}}}$ and $\boxed{b = \mu_0 + z_{\alpha/2}\dfrac{\sigma_X}{\sqrt{n}}}$.

So we reject H_0 iff $|z| > z_{\alpha/2}$. By symmetry, p-value $= \mathbb{P}(|Z| > |z|) = 2\,\mathbb{P}(Z > |z|)$.

z-test for μ_X (σ_X^2 known)

Let $z = \dfrac{\overline{x} - \mu_0}{\sigma_X/\sqrt{n}}$ be the observed value of the test statistic. Reject H_0 at level α when

- right test ($H_1 : \mu_X > \mu_0$): $z > z_\alpha$, with p-value $\mathbb{P}(Z > z)$;
- left test ($H_1 : \mu_X < \mu_0$): $z < -z_\alpha$, with p-value $\mathbb{P}(Z < z)$;
- two-sided test ($H_1 : \mu_X \neq \mu_0$): $|z| > z_{\alpha/2}$, with p-value $2\,\mathbb{P}(Z > |z|)$.



Shaded tails are rejection regions; the remaining central area has probability $1-\alpha$.

Numerical illustration: σ_X^2 known

Let $X_1, \ldots, X_{25}$ be a random sample from $N(\mu_X, \sigma_X^2)$ with $\sigma_X = 0.16$, and suppose the observed sample mean is $\overline{x} = 8.091$. Test $H_0 : \mu_X = 8$ against $H_1 : \mu_X \neq 8$ at the 0.01 level of significance.

Critical-value approach: Since $\alpha = 0.01$, $z_{\alpha/2} = z_{0.005} = 2.5758$, and $\dfrac{\overline{x} - \mu_0}{\sigma_X/\sqrt{n}} = \dfrac{8.091 - 8}{0.16/\sqrt{25}} = \dfrac{0.091}{0.032} = 2.84375 > 2.5758$, so the null hypothesis is rejected.

p-value approach: p-value $= 2\,\mathbb{P}(Z > 2.84375) = 2(1 - 0.9977707) \approx 0.0045 < 0.01$, so the null hypothesis is rejected. The two approaches agree, and a p-value well below α means H_0 is strongly rejected.

If instead $\overline{x} = 8.051$, the test statistic is only $1.59 < 2.5758$ and H_0 is not rejected. Increasing σ_X or reducing n also shrinks the test statistic, making rejection harder.

5.2.2 Tests for μ_X: σ_X^2 Unknown

When σ_X^2 is unknown we replace it by S_{n-1}^2. By the theorem above, under $H_0 : \mu_X = \mu_0$ the test statistic is $T_{n-1} = \dfrac{\overline{X} - \mu_0}{S_{n-1}/\sqrt{n}} \sim t(n-1)$, where the observed value is the t **value** $t = \dfrac{\overline{x} - \mu_0}{s_{n-1}/\sqrt{n}}$.

The three tests are derived exactly as in the known-variance case, with z_α replaced by $t_{n-1,\alpha}$ and the observed s_{n-1} replacing the known σ_X.

One-sided right test. For $H_1 : \mu_X > \mu_0$ reject H_0 if $\overline{x} > c$, and $\alpha = \mathbb{P}(\overline{X} > c \mid \mu_X = \mu_0) = \mathbb{P}\left(T_{n-1} > \dfrac{c - \mu_0}{s_{n-1}/\sqrt{n}}\right)$, so $\dfrac{c - \mu_0}{s_{n-1}/\sqrt{n}} = t_{n-1,\alpha}$, i.e. $\boxed{c = \mu_0 + t_{n-1,\alpha}\dfrac{s_{n-1}}{\sqrt{n}}}$.

So reject H_0 iff $\overline{x} > \mu_0 + t_{n-1,\alpha}\dfrac{s_{n-1}}{\sqrt{n}}$, i.e. iff $t > t_{n-1,\alpha}$, with p-value $\mathbb{P}(T_{n-1} > t)$.

One-sided left test. For $H_1 : \mu_X < \mu_0$ reject H_0 if $\overline{x} < c$, and $\alpha = \mathbb{P}(\overline{X} < c \mid \mu_X = \mu_0) \implies \dfrac{c - \mu_0}{s_{n-1}/\sqrt{n}} = -t_{n-1,\alpha}$, i.e. $\boxed{c = \mu_0 - t_{n-1,\alpha}\dfrac{s_{n-1}}{\sqrt{n}}}$.

So reject H_0 iff $t < -t_{n-1,\alpha}$, with p-value $\mathbb{P}(T_{n-1} < t)$.

Two-sided test. For $H_1 : \mu_X \neq \mu_0$ reject H_0 if $\overline{x} < a$ or $\overline{x} > b$, with $\mathbb{P}(\overline{X} < a \mid \mu_X = \mu_0) + \mathbb{P}(\overline{X} > b \mid \mu_X = \mu_0) = \alpha$, which gives $\boxed{a = \mu_0 - t_{n-1,\alpha/2}\dfrac{s_{n-1}}{\sqrt{n}}}$ and $\boxed{b = \mu_0 + t_{n-1,\alpha/2}\dfrac{s_{n-1}}{\sqrt{n}}}$.

So reject H_0 iff $|t| > t_{n-1,\alpha/2}$, with p-value $\mathbb{P}(|T_{n-1}| > |t|) = 2\,\mathbb{P}(T_{n-1} > |t|)$.

t-test for μ_X (σ_X^2 unknown)

Let $t = \dfrac{\overline{x} - \mu_0}{s_{n-1}/\sqrt{n}}$ be the observed t value. Reject H_0 at level α when

- right test ($H_1 : \mu_X > \mu_0$): $t > t_{n-1,\alpha}$, with p-value $\mathbb{P}(T_{n-1} > t)$;
- left test ($H_1 : \mu_X < \mu_0$): $t < -t_{n-1,\alpha}$, with p-value $\mathbb{P}(T_{n-1} < t)$;
- two-sided test ($H_1 : \mu_X \neq \mu_0$): $|t| > t_{n-1,\alpha/2}$, with p-value $2\,\mathbb{P}(T_{n-1} > |t|)$.

```
# R corner: elephant call frequencies (Hz), n = 12
freq <- c(14, 16, 17, 17, 24, 20, 32, 18, 29, 31, 15, 35)
t.test(freq, mu = 10)   # H0: mu = 10 (two-sided)
#   t = 5.6877, df = 11, p-value = 0.00014 -> reject H0
t.test(freq, mu = 20)   # t = 1.0761, p-value = 0.3049
t.test(freq, mu = 23)   # t = -0.3074, p-value = 0.7642
qt(0.975, df = 11)      # 2.200985, the critical value
```

Here $\overline{x} = 268/12 = 22.33$ and $s_{n-1} \approx 7.51$, so the t value for $\mu_0 = 10$ is $12.33/(7.51/\sqrt{12}) \approx 5.69$. R reports only this t value; for a two-sided test we take its absolute value and compare with $t_{11,0.025} = 2.201$, which gives rejection. For $\mu_0 = 20$ and $\mu_0 = 23$ the t values are small, so H_0 is not rejected — consistent with the fact that both 20 and 23 lie inside the 95% confidence interval $[17.56, 27.11]$ for μ_X obtained in Section 4.3 (the confidence-interval and hypothesis-testing approaches agree).

5.2.3 Worked Example: One-Sided Test and Power

The average length of time for students to register for classes has been 46 minutes. A new procedure is tried: a random sample of $n = 12$ students gives $\overline{x} = 42$ minutes with $s_{n-1} = 11.9$ minutes. Assume the data come from a normal distribution.

(i) **Test** $H_0 : \mu_X = 46$ against $H_1 : \mu_X < 46$ at $\alpha = 0.05$. Since σ_X^2 is unknown this is a one-sided left t-test with df $= 11$. We reject H_0 if $\overline{x} < \mu_0 - t_{n-1,\alpha}\dfrac{s_{n-1}}{\sqrt{n}} = 46 - 1.7959 \cdot \dfrac{11.9}{\sqrt{12}} = 46 - 6.169 = 39.831$.

Since $\overline{x} = 42 > 39.831$ (equivalently, the t value is $-1.164 > -1.7959$), we do not reject H_0: there is not enough evidence that the mean registration time is now less than 46 minutes.

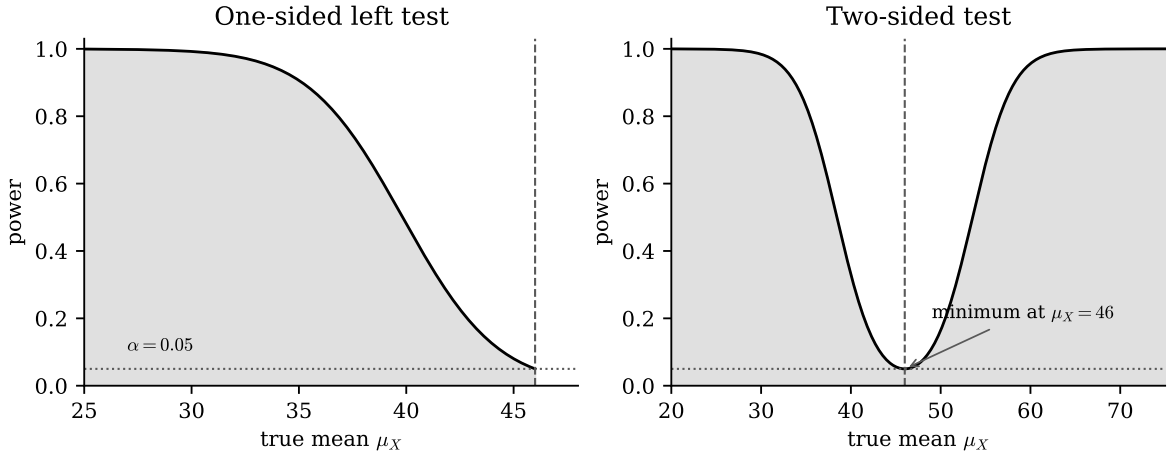
(ii) **Power of the one-sided left test.** The power (or the Type II error probability β) is a function of μ_X for $\mu_X < 46$: $\text{power}(\mu_X) = \mathbb{P}\left(\overline{X} < 46 - t_{11,0.05}\dfrac{11.9}{\sqrt{12}} \,\middle|\, \mu_X\right) = \mathbb{P}\left(T_{11} < \dfrac{39.83072 - \mu_X}{11.9/\sqrt{12}}\right)$.

Equivalently, $\text{power}(\mu_X) \approx \Phi\left(\dfrac{39.83072 - \mu_X}{\sigma_X/\sqrt{12}}\right)$. Since Φ is increasing while $39.83072 - \mu_X$ increases as μ_X decreases, the power increases as μ_X decreases: the further the true mean lies below 46, the easier the deviation is to detect.

```
# R corner: power curve of the one-sided left test
ux <- seq(0, 45.9, by = 0.3)
power <- rep(0, length(ux))
for (i in 1:length(ux)) {
  t <- (39.83072 - ux[i]) / (11.9/sqrt(12))
  power[i] <- pt(t, df = 11)
}
plot(ux, power, xlab = expression(mu[X]))
```

(iii) **Power of the two-sided test** $H_0 : \mu_X = 46$ against $H_1 : \mu_X \neq 46$. With $t_{11,0.025} = 2.201$ the two critical values are $46 \pm 2.201 \cdot (11.9/\sqrt{12})$, i.e., $a = 38.4391$ and $b = 53.5609$, so $\text{power}(\mu_X) = \mathbb{P}\left(T_{11} < \dfrac{a-\mu_X}{11.9/\sqrt{12}}\right) + \mathbb{P}\left(T_{11} > \dfrac{b-\mu_X}{11.9/\sqrt{12}}\right)$. This power is minimized at $\mu_X = 46$ and increases as μ_X moves away from 46 in either direction. Replacing T_{11} by $N(0,1)$ for large n, $\text{power}(\mu_X) \approx \Phi\left(\dfrac{a-\mu_X}{\sigma_X/\sqrt{n}}\right) + 1 - \Phi\left(\dfrac{b-\mu_X}{\sigma_X/\sqrt{n}}\right)$, and differentiating, $\dfrac{d}{d\mu_X}\text{power}(\mu_X) \approx \dfrac{1}{\sigma_X/\sqrt{n}}\left[\varphi\left(\dfrac{b-\mu_X}{\sigma_X/\sqrt{n}}\right) - \varphi\left(\dfrac{a-\mu_X}{\sigma_X/\sqrt{n}}\right)\right]$, which vanishes when $|b-\mu_X| = |a-\mu_X|$, i.e. at $\mu_X = (a+b)/2 = 46$, and is positive (resp. negative) for $\mu_X > 46$ (resp. $\mu_X < 46$); hence $\mu_X = 46$ is the minimum.

```
# R corner: power curve of the two-sided test
ux <- seq(0, 100, by = 0.3)
power <- rep(0, length(ux))
for (i in 1:length(ux)) {
  ta <- (38.4391 - ux[i]) / (11.9/sqrt(12))
  tb <- (53.5609 - ux[i]) / (11.9/sqrt(12))
  power[i] <- pt(ta, df = 11) + 1 - pt(tb, df = 11)
}
plot(ux, power, xlab = expression(mu[X]))
abline(v = 46, lty = 2, lwd = 1)
```



5.2.4 Tests for σ_X^2: μ_X Unknown

Under $H_0 : \sigma_X^2 = \sigma_0^2$ the statistic $U_{n-1} = (n-1)S_{n-1}^2/\sigma_0^2 \sim \chi^2(n-1)$ by the theorem at the start of 5.2, so S_{n-1}^2 is the test statistic, with observed value $u = \dfrac{(n-1)s_{n-1}^2}{\sigma_0^2}$.

One-sided right test. For $H_1 : \sigma_X^2 > \sigma_0^2$ reject H_0 if $s_{n-1}^2 > c$. Under H_0, $U_{n-1} \sim \chi^2(n-1)$, so $\alpha = \mathbb{P}(S_{n-1}^2 > c \mid \sigma_X^2 = \sigma_0^2) = \mathbb{P}\left(U_{n-1} > \dfrac{(n-1)c}{\sigma_0^2}\right)$, i.e. $\boxed{c = \dfrac{\sigma_0^2 \chi^2_{n-1,\alpha}}{n-1}}$.

Hence reject H_0 iff $u > \chi^2_{n-1,\alpha}$, with p-value $\mathbb{P}(U_{n-1} > u)$.

One-sided left test. For $H_1 : \sigma_X^2 < \sigma_0^2$ reject H_0 if $s_{n-1}^2 < c$, and $\alpha = \mathbb{P}(S_{n-1}^2 < c \mid$

$\sigma_X^2 = \sigma_0^2) \implies \dfrac{(n-1)c}{\sigma_0^2} = \chi^2_{n-1,1-\alpha}$, i.e. $\boxed{c = \dfrac{\sigma_0^2 \chi^2_{n-1,1-\alpha}}{n-1}}$.

Hence reject H_0 iff $u < \chi^2_{n-1,1-\alpha}$, with p-value $\mathbb{P}(U_{n-1} < u)$.

Two-sided test. For $H_1 : \sigma_X^2 \neq \sigma_0^2$ reject H_0 if $s^2_{n-1} < a$ or $s^2_{n-1} > b$, splitting α equally: $\mathbb{P}(S^2_{n-1} < a \mid \sigma_X^2 = \sigma_0^2) + \mathbb{P}(S^2_{n-1} > b \mid \sigma_X^2 = \sigma_0^2) = \alpha$, giving $\boxed{a = \dfrac{\sigma_0^2 \chi^2_{n-1,1-\alpha/2}}{n-1}}$ and $\boxed{b = \dfrac{\sigma_0^2 \chi^2_{n-1,\alpha/2}}{n-1}}$.

Hence reject H_0 iff $u < \chi^2_{n-1,1-\alpha/2}$ or $u > \chi^2_{n-1,\alpha/2}$. Because the χ^2 distribution is asymmetric, the p-value is twice the smaller tail probability, $2\min\{\mathbb{P}(U_{n-1} < u),\ \mathbb{P}(U_{n-1} > u)\}$.

χ^2-test for σ_X^2 (μ_X unknown)

Let $u = \dfrac{(n-1)s^2_{n-1}}{\sigma_0^2}$ be the observed value of the test statistic. Reject H_0 at level α when

- right test ($H_1 : \sigma_X^2 > \sigma_0^2$): $u > \chi^2_{n-1,\alpha}$, with p-value $\mathbb{P}(U_{n-1} > u)$;
- left test ($H_1 : \sigma_X^2 < \sigma_0^2$): $u < \chi^2_{n-1,1-\alpha}$, with p-value $\mathbb{P}(U_{n-1} < u)$;
- two-sided test ($H_1 : \sigma_X^2 \neq \sigma_0^2$): $u < \chi^2_{n-1,1-\alpha/2}$ or $u > \chi^2_{n-1,\alpha/2}$, with p-value $2\min\{\mathbb{P}(U_{n-1} < u),\ \mathbb{P}(U_{n-1} > u)\}$.

Battery life: test for σ_X

A manufacturer claims that the life of his batteries is normally distributed with standard deviation 0.9 year. A random sample of 10 batteries has standard deviation 1.2 years. Test $H_0 : \sigma_X = 0.9$ against $H_1 : \sigma_X > 0.9$ at $\alpha = 0.05$; here μ_X is unknown, so df $= 9$.

With $n = 10$ and $s_{n-1} = 1.2$, $\dfrac{(n-1)s^2_{n-1}}{\sigma_0^2} = \dfrac{9(1.2)^2}{(0.9)^2} = \dfrac{12.96}{0.81} = 16.0$. Since $\chi^2_{9,0.05} = 16.919$ and $16.0 < 16.919$, we do not reject H_0. Equivalently, the p-value is $\mathbb{P}(\chi^2(9) > 16.0) \approx 0.067 > 0.05$. There is not enough evidence to conclude that $\sigma_X > 0.9$ year. In R, `pchisq(16, 9, lower.tail = FALSE)` ≈ 0.0669 and `qchisq(0.95, 9)` $= 16.919$.

5.2.5 Summary of the Tests

Parameter	Assumption	Test statistic under H_0
μ_X	σ_X^2 known	$\dfrac{\overline{X} - \mu_0}{\sigma_X/\sqrt{n}} \sim N(0,1)$
μ_X	σ_X^2 unknown	$\dfrac{\overline{X} - \mu_0}{S_{n-1}/\sqrt{n}} \sim t(n-1)$
σ_X^2	μ_X unknown	$\dfrac{(n-1)S^2_{n-1}}{\sigma_0^2} \sim \chi^2(n-1)$

Reject H_0 when the statistic lies in the corresponding tail. ■